

The Spanish PYTA Morphome Dissolved

Draft 2 (Comments Welcome!)

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Abstract: Many Romance languages exhibit a morphomic pattern dubbed PYTA (for *{perfecto/pretérito} y tiempos afines*—see especially Maiden 2018, Ch 4). Spanish exhibits a striking instance of this phenomenon. No matter how irregular a Spanish verb is otherwise, it always has the same stem for both imperfect subjunctive paradigms and for the future subjunctive (which is defunct in the modern spoken language, but was in active use a few centuries ago), and this stem can be unerringly arrived at by taking the 3rd person plural preterite indicative form and subtracting the sequence *-ron*. This generalization is exceptionless, but it is not directly formulable in a piece-based view of morphology. For this reason, the PYTA morphome has been suggested to require a paradigm-based view of morphology (see for instance O’Neill 2014 and references cited there). In this paper, I will argue that the PYTA morphome, at least as it arises in Spanish, is actually the result of two independent generalizations interacting. One of these generalizations involves suppletion conditioned by a morphosyntactic feature —Present (usually in combination with the feature +Finite), the other is phonological and concerns (i) the conditioning of vowel height alternations in certain verb roots of the 3rd conjugation, and (ii) the circumstances under which the Theme Vowel of 2nd and 3rd conjugation verbs diphthongizes. Once these generalizations are independently explained (as they must be, since they each affect different parts of the paradigm which extend beyond the claimed morphomic pattern), there is no work left for a direct statement of the morphomic pattern to do, and thus no threat to piece-based models of morphology from this quarter.

1 Introduction: Two Big Questions and How Morphomes Bear on Them¹

Realizational theories of morphology are united by their commitment to the existence of Rules of Exponence, and by the idea that such rules compete with each other via some version of the Elsewhere Condition (Pāṇini’s Principle). They are divided on at least the following two Big Questions.

- (1) **A Big Question (about pieces)**
Is the input to the Rules of Exponence (a) a syntactic (or at least syntax-like) hierarchical structure, as on piece-based views; or (b) an unstructured bag of features, as on word-based views?
- (2) **Another Big Question (about paradigms)**
Are Rules of Exponence (a) sensitive only to information in the structure they are exponing, or are they also (b) potentially sensitive to the way other structures are realized, via transderivational principles that regulate form correspondences?

Of the arguments in favor of the (b) answers to these Big Questions, so-called *Priscianic* (Matthews 1972) or *Morphomic* (Aronoff 1994) patterns stand pre-eminent (see especially Maiden 2018, 2021; O’Neill 2014, 2015, 2024; and references cited in these works for documentation of such patterns and for arguments that they favor word-and-paradigm theories of morphology). Many famous examples of morphomes from Aronoff’s original discussion, including English participles, the Latin *t*-stem, and the Semitic root, involve a single position of exponence with several allomorphs or special morphophonology, coupled with more than one meaning or syntactic function. This sort of morphome is not, in fact, especially problematic in a piece-based theory with “Late Insertion at Both Interfaces”—that is, one that has Rules of Semantic Interpretation which relate syntactic primes to their denotations, analogous (but orthogonal) to the Rules of Exponence (see Marantz & Myler to appear and references cited there). In such a theory, pieces may, but need not, have a single meaning—all that is criterial for being a piece is having a statable (even if complex) morphosyntactic distribution, and it seems to me that English participles, the Latin *t*-stem, and the Semitic root all meet this criterion. Let us call morphomes of this kind, which are expected even on a piece-based view given the Autonomy of (Morpho)Syntax, *Piece-based Morphomes*.

Much more troubling are what I will call *Process-based Morphomes*. These are morphomic patterns which are not traceable to a single position of exponence, but rather seem to require a statement in terms of applying some procedure to a form from one part of the paradigm to derive the input to rules for another part of the paradigm. This paper focuses on an especially striking case of this kind: the so-called PYTA Morphome of Romance languages (Maiden 2018, Ch4), and specifically its manifestation in Spanish. PYTA, which stands for *Perfecto/Pretérito y tiempos afines*, can be characterized as follows:

¹Acknowledgements here. All errors are mine.

(3) The Spanish PYTA Morphome

The stem for the following paradigms can be derived by taking the third person plural preterite form of a given verb, and subtracting the sequence *-ron*:

- a. The imperfect subjunctive in *-ra*
- b. The imperfect subjunctive in *-se*
- c. The future subjunctive (obsolete in modern spoken Spanish)²

This generalization is completely exceptionless. No matter how irregular a Spanish verb is otherwise, its imperfect subjunctive and future subjunctive forms can always be accurately arrived at by following this procedure, a fact which L2 Spanish pedagogy often exploits by explicitly teaching the pattern to learners (see for example Gordon & Stillman 2004, 23). The table below contains something of a Rogue’s Gallery of irregular Spanish verbs by way of illustration (as well as *dormir* and *pedir*, whose properties, while not entirely predictable, nonetheless fall into a broader class).

(4) Imperf and Fut Subjunctive Stems from the 3rd Plural Preterit Minus *-ron*

Verb	3rd Person Plural Preterite	Imperfect/Future Subjunctive Stem
<i>ser</i> ‘to be ₁ ’	<i>fu</i> eron	<i>fue</i> -
<i>estar</i> ‘to be ₂ ’	<i>estuv</i> ieron	<i>estuvie</i> -
<i>dar</i> ‘to give’	<i>dier</i> on	<i>die</i> -
<i>poner</i> ‘to put’	<i>pus</i> ieron	<i>pusie</i> -
<i>decir</i> ‘to say’	<i>dij</i> eron	<i>dije</i> -
<i>querer</i> ‘to want’	<i>quis</i> ieron	<i>quisie</i> -
<i>saber</i> ‘to know’	<i>sup</i> ieron	<i>supie</i> -
<i>dormir</i> ‘to sleep’	<i>durm</i> ieron	<i>durmie</i> -
<i>pedir</i> ‘to request’	<i>pid</i> ieron	<i>pidie</i> -

Now, there is no clear sense in which the imperfect subjunctive or the future subjunctive contains the meaning or the morphosyntactic features of the third person plural preterite “minus *-ron*”, whatever these would amount to. Moreover, the subtracted portion *-ron* corresponds to at least two pieces, rather than one. Both of these facts make the pattern difficult to state in a piece-based view. Another observation is that the procedure in (3) is inherently *transderivational*: if one is spelling out the 1st person singular future subjunctive of some verb, one can’t calculate the form from these features alone, according to (3). One must in addition look across to a different cell in the paradigm and perform an operation on the contents of that cell.

Those of us who believe in the (a) answers to the Two Big Questions I started with (in other words, those who believe in a piece-based, strict-generation theory of grammar) are therefore on the hook for an alternative explanation of such apparent cases of Process-based Morphemes. The rest of this paper provides such an explanation for the case of PYTA in Spanish. Section 2 begins by giving a more detailed overview of the pattern itself, followed by a sketch of my proposed solution, which consists of a morphosyntactic plank and a morphophonological plank. Section 3 is devoted to explaining the morphosyntactic plank in more detail, with the key role being played by a feature –Pres(ent) (often in combination with a more familiar feature +Fin(ite)). Section 4

²I include the future subjunctive forms as one of the explananda here, despite the fact that they are no longer in common use. The reason for this is that these forms *were* in common use a few centuries ago, and their equivalents are still alive and kicking in Portuguese. I therefore take it that Spanish-with-the-future-subjunctive is a possible state of the language faculty, which grammatical theory must confront.

then does likewise for the morphophonological plank, which makes crucial use of Logical Phonology (Hale & Reiss 2008 et seq.) as a theory of the Melodic Tier, and of Strict CV (Lowenstamm 1996; Scheer 2004, many others) as a theory of the Timing Tier. Section 5 provides a summary and concludes the paper by considering some broader theoretical implications of this discussion. I will argue that all linguists must bite at least one of the following three bullets with respect to morphophonology: transderivationality, anti-modularity, or abstractness. Section 6 consists of appendices, which bring together the rules and structures introduced across the discussion, plus some others not covered in the body of the paper (for expository and space-related reasons).

2 More on the Data and a Sketch of the Solution

To provide the reader with a more complete sense of the patterns involved, the tables in (5) and (6) give the full paradigms of the irregular 2nd conjugation verb *querer* ‘to want’, and of the semi-irregular 3rd conjugation verb *pedir*. The cells corresponding to the PYTA Morphome are shaded in light blue; the significance of the underlined and bolded parts of some of the forms is explained below.

(5) *querer* ‘to want’

INDICATIVE:			
PRESENT	IMPERFECT	PRETERITE	FUTURE
quiero	quería	quise	querré
quieres	querías	quisiste	querrás
quiere	quería	quiso	querrá
queremos	queríamos	quisimos	querremos
queréis	queríais	quisisteis	querréis
quieren	querían	quisieron	querrán
CONDITIONAL:			
querría			
querrías			
querría			
querríamos			
querríais			
querrían			
SUBJUNCTIVE:			
PRESENT	IMPERFECT-1	IMPERFECT-2	FUTURE
quiera	quisiera	quisiese	quisiere
quieras	quisieras	quisieses	quisieres
quiera	quisiera	quisiese	quisiere
queramos	quisiéramos	quisiésemos	quisiéremos
queráis	quisierais	quisieseis	quisiereis
quieran	quisieran	quisiesen	quisieren
IMPERATIVE:			
quiere			
quered			
INFINITIVE:	GERUND:	PAST PARTICIPLE:	
querer	queriendo	querido	

(6) *pedir* ‘to request’

INDICATIVE:			
PRESENT	IMPERFECT	PRETERITE	FUTURE
pid o pid es pid e pedimos pedís pid en	pedía pedías pedía pedíamos pedíais pedían	pedí pediste pid ió pedimos pedisteis pid ieron	pediré pedirás pedirá pediremos pediréis pedirán
CONDITIONAL:			
pediría pedirías pediría pediríamos pediríais pedirían			
SUBJUNCTIVE:			
PRESENT	IMPERFECT-1	IMPERFECT-2	FUTURE
pid a pid as pid a pid amos pid áis pid an	pid iera pid ieras pid iera pid iéramos pid ierais pid ieran	pid iese pid ieses pid iese pid iésemos pid ieseis pid iesen	pid iere pid ieres pid iere pid iéremos pid iereis pid ieren
IMPERATIVE:			
pid e pedid			
INFINITIVE:	GERUND:	PAST PARTICIPLE:	
pedir	pid iendo	pedido	

My proposal, in outline, is this. Irregular stem allomorphs such as the bolded element *quis-* in (5) are conditioned by the feature –Pres, in many cases (as here) in combination with +Fin; that is, these allomorphs are selected in non-present finite environments.³ Vowel alternations in the Theme Vowel for 2nd and 3rd conjugation verbs, including crucially the underlined diphthongal forms, are instead a matter of phonology, and the same is true of certain vowel height alternations in the roots of some 3rd conjugation verbs (including in the bolded stem *pid-* in (6); here I adapt the account in Brame & Bordelois 1973, 1974). Crucially, both the morphosyntactically conditioned allomorphy and the vowel alternations have to be dealt with independently of the PYTA Morpheme; as scrutiny of (5) and (6) will show, each of the underlined and bolded elements extends in different ways outside of the blue shaded cells corresponding to the PYTA pattern. Irregular stem allomorphs are found throughout the preterite (not just in the 3rd person plural), the high alternant of alternating vowels in 3rd conjugation verbs is found in a great many parts of the paradigm, and the diphthongal form of the Theme Vowel is found in the gerund as well as within the PYTA pattern. The punchline will be that, once one has said what one needs to say to

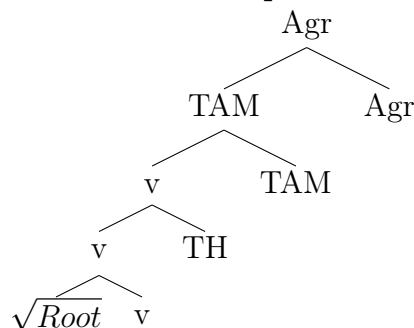
³See Bembridge (2022) and section 3 below on the apparent difficulties for this way of putting things posed by the imperfect indicative, the future indicative, and the conditional.

cover the distribution of the suppletive stem allomorphs and the various relevant vowel alternants, **there is no work left for a morphomic statement to do**. Let us now begin to flesh out the details, beginning with the morphosyntax.

3 The Details Part 1: Morphosyntax and Special Rules tied to $-Pres$, $+Fin$

Following much earlier work, and especially Arregi (2000) and Roca (2010), I assume that the clausal morphosyntax of Spanish builds a complex head with at least the internal structure depicted in (7), where ‘v’ is the verbalizer, TH stands for the Theme Vowel, TAM stands for a Tense/Aspect/Mood head, and Agr stands for Agreement morphology (for the person and number of the subject). Against the background of this structure, the suppletive allomorphy in the stem of *querer* can be handled by the rules in (8); other verbs that exhibit suppletion in PYTA forms can be assigned a similar account. Note that, here and throughout, I treat irregular stem allomorphy in the preterite and related forms as involving *root* suppletion. While I do think that suppletion is involved, it seems to me that the patterns internal to these irregulars point to the actual suppletion being in the realization of ‘v’. In other work in progress I pursue this possibility, but I have abstracted away from it here for the sake of simplicity.

(7) **The Syntactic Structure of the Spanish Verb**



(8) **Rules of Exponence for the root of the verb *querer* ‘to want’**

- a. $\sqrt{want} \leftrightarrow \text{kis} / __ \dots [-Pres, +Fin]_{TAM}$ ⁴
- b. $\sqrt{want} \leftrightarrow \text{ke}^{CV} \text{r}$ ⁵

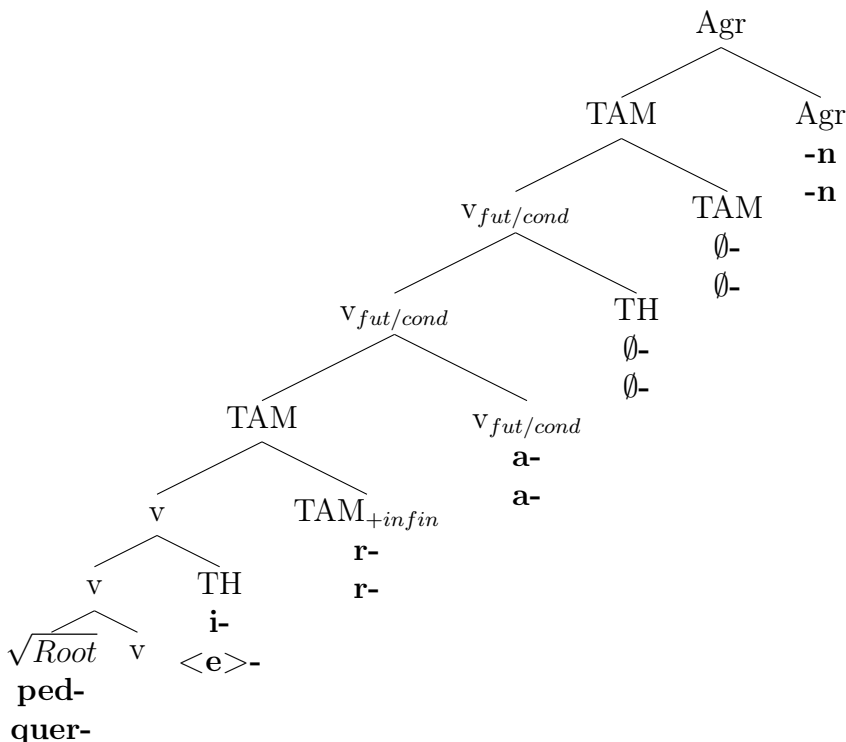
⁴The ellipses in the conditioning environment for this rule indicate that material may intervene between the root and the conditioning TAM morpheme—in particular, the Theme Vowel. It seems to me that the present data present us with a bullet regarding the locality of allomorphy which, unfortunately, we have no choice but to bite, *pace* Calabrese (2015), Embick (2012); compare also Bembridge (2022, p.161) on these Spanish facts in particular, and on the more general issue Moskal & Smith (2016); Bešlin (2025). Theories of the locality of allomorphy in which it is conditioned by a version of Rizzi’s (1990) Relativized Minimality (see, in the context of Spanish allomorphy specifically, Bembridge 2022, p.167), by Cyclic Locality, or by Structural Adjacency, can all handle the relevant dependency (the latter only if my suspicion that the real suppletion here pertains to the verbalizer, rather than the root, turns out to be correct). However, theories based on Linear Adjacency turn out to be too restrictive—there is no escaping the need to state an allomorphic dependency across the Theme Vowel in Spanish verbs.

⁵The superscript ‘CV’ here indicates that the associated vowel undergoes diphthongization when stressed, giving rise to forms like the first person singular present indicative *quiero*. Such alternations are taken up again in section 4.3.

For the imperfect indicative, I propose that it is subject to the following Impoverishment Rule deleting the $-Pres$ feature:

- While (9) is of course a stipulation, it is a highly motivated one. The imperfect indicative is unique amongst past tenses in the language, in that it is unusually *regular*. Only two verbs, *ser* ‘to be₁’ and *ir* ‘to go’, have irregular imperfect indicative forms (this is not the only way in which these two verbs pattern together in the language; see Fábregas 2016; Kayne 2020). The Impoverishment Rule in (9) captures the imperfect indicative’s uniqueness in a way that is linked to its unusual regularity (since the feature deleted is one that, by hypothesis, conditions a number of irregularities in the language).

(10) *pedirán* ‘they will request’/*querrán* ‘they will want’⁶



7

The idea is that Spanish has no TAM morpheme in its inventory of pieces corresponding to the future indicative or the conditional. Instead, forms with such meanings must be built by merging a special ‘v’ head with the infinitive. This head, which I label $v_{fut/cond}$, has the lexical entry in (11). In this and other lexical entries, a filled bullet point indicates a Merge operation after which the lexical item in question is to project, and a clear bullet point indicates a Merge operation after which the other syntactic object merged with is to project (adapting the conventions of Müller 2009, 2011, 2013; also Zyman Under revision for the clear bullet in particular). Note that, here and elsewhere, I assume that Theme Vowels are introduced into the syntactic structure through Merge, and that inflectional class features themselves are also present in the syntax, following Collins & Kayne (2023), Myler (2025), and Zyman (2025/submitted), but nothing hinges on this.

(11) **Future/Conditional Builder**

$$\left[\begin{array}{l} \text{CAT: } v, \text{ fut/cond} \\ \text{CLASS: } +\alpha, +\beta \\ \bullet \text{TAM}_{+infin} \\ \bullet \text{TH} \\ \circ \text{TAM}_{-perf, -sbjv} \end{array} \right] v_{fut/cond}$$

The instruction to merge with a TAM morpheme bearing the features $[-\text{Perf}, -\text{Sbjv}]$ ensures that the higher TAM marker is either present indicative or imperfect indicative. If the present indicative TAM head is merged, we get what is known as the future indicative form; if the imperfect indicative TAM head is merged, we instead get what is known as the conditional form. I follow Iatridou (2000), Arregui (2005), and many others in proposing that $v_{fut/cond}$ introduces an underspecified modal semantics which interacts with the meaning of the present and of the imperfect to yield the meanings of the future indicative and conditional respectively. The fact that $v_{fut/cond}$ bears the inflectional class features of the 2nd conjugation $[+\alpha, +\beta]$ (Oltra-Massuet 1999; Arregi 2000), will, given the rules introduced below in Section 4, account for the appearance of the Theme Vowel /i/ in the conditional endings, and for the appearance of the vowel /e/ in a subset of the future indicative forms. The item $v_{fut/cond}$ itself is subject to the following Rules of Exponence:

- (12) a. $v_{fut/cond} \leftrightarrow (a / _ \dots [+Pres]_{TAM} [-\text{author}, -\text{pl}]_{AGR}$
b. $v_{fut/cond} \leftrightarrow (a / _ \dots [+Pres]_{TAM} [-\text{participant}, +\text{pl}]_{AGR}$
c. $v_{fut/cond} \leftrightarrow (CV$

The parens ‘(’ in these rules indicates that these forms project a left bracket to Line 0 of the metrical grid, an assumption that will, in combination with the rules of Accent Assignment presented later, account for the fact that stress surfaces on the ending in the relevant forms. The Elsewhere allomorph in (12c) is an empty CV sequence which will give the Theme Vowel a place to dock (I will propose in section 4 that Theme Vowels are not pre-linked to the Timing Tier). The other two allomorphs are accidental homophones, both consisting of a single /a/ which is prelinked to the Timing Tier, and with no additional timing slots. When these allomorphs are chosen, the Theme Vowel will have nowhere to dock, and only the /a/ will surface. Thus are the forms in the future indicative paradigms explained (see (5) and (6)).

Whether the analyses of the semantics and the allomorphy of $v_{fut/cond}$ just sketched will turn out to be the right ones is a question I will have to leave open.⁷ The most important thing here is something that will follow from any version of this analysis: there will be no TAM marker local to the root in these structures that bears the features $[-Pres, +Fin]$. With respect to the rules in (8), this means that we will correctly predict that the $/ke^{CV}r/$ allomorph will win the competition by default in the future indicative and the conditional.

With this, the first generalization that gives rise to the PYTA Morpheme is explained: relevant suppletive allomorphs are conditioned by the features $[-Pres, +Fin]$. We now turn to the phonological generalizations that cover the rest of this pattern.

4 The Details Part 2: (Theme) Vowel Alternations

This part of the discussion begins in section 4.1 by overviewing the core data on non-automatic vowel alternations in Spanish, and by sketching my proposals regarding the Underlying Representations of alternating and non-alternating vowels. Examples of the Rules of Exponence responsible for introducing such Underlying Representations are also provided. Section 4.2 details the specific theoretical proposals I adopt regarding the nature of phonological representations, dealing in turn with the Melodic Tier and the Timing Tier. This serves as background for section 4.3, where a specific phonological rule system covering most of the non-automatic vowel alternations in the language is presented, along with sample derivations. Section 4.4 concludes by summarizing, in one place, the specific subparts of this rule system that explain the vowel-related aspects of PYTA as a special case.

4.1 The Data and Proposed Rules of Exponence

The PYTA Morpheme holds trivially in the 1st conjugation (the *-ar* verbs): the root and the Theme Vowel never exhibit special allomorphy in the relevant cells, as illustrated over the page in (14). The Theme Vowel allomorphy that is relevant for present purposes all arises in the 2nd and 3rd conjugations (respectively, the verbs with infinitives in *-er* and *-ir*). As it happens, however, the distinction between 2nd and 3rd conjugation Theme Vowels collapses in most TAM combinations (in favor of the 3rd conjugation, it would seem). The relevant syncretism patterns reveal themselves in a comparison of the yellow shaded cells in the paradigms provided on subsequent pages as (17) and (18).

As anticipated in the discussion of $v_{fut/cond}$ in section 3, I assume that roots⁸ and some verbalizers bear class features in the syntax (Myler 2025; Zyman 2025/submitted). Following Oltra-Massuet (1999) and Arregi (2000), I give these features the arbitrary designations $[+/-\alpha]$, $[+/-\beta]$ (see Davis 1991, p.151 for a similar idea), with the conjugation classes themselves defined in terms of the following feature specifications.

⁷An alternative on the allomorphy side would be to have (12c) be the only Rule of Exponence that applies to $v_{fut/cond}$, and instead have Impoverishment Rules deleting conjugation class features in the same environments that (12a) and (12b) apply to. The $/a/$ in 2nd singular, 3rd singular, and 3rd plural future indicative forms would then be the default Theme Vowel allomorph, rather than a realization of $v_{fut/cond}$ itself.

⁸Alternatively, roots might never bear class features themselves, as argued by Acquaviva (2009). It could be that only verbalizers carry conjugation class features, and that the pairing up of roots and verbalizers is achieved via selection (Merchant 2019) or some notion of ‘licensing conditions’ for roots (see Kramer 2015, pp.49-57 and references cited there on the latter).

(13) **Feature Specifications for Spanish Conjugation Classes**

- a. 1st Conjugation: $[-\alpha]$
- b. 2nd Conjugation: $[+\alpha, +\beta]$
- c. 3rd Conjugation: $[+\alpha]$

All verbalizers merge first with their host (the root in the cases we are discussing), and secondly with a Theme Vowel node TH. TH undergoes Agree with the nearest root/verbalizer in its c-command domain for its class features.

(14) **1st Conjugation Verb *cantar* ‘to sing’**

INDICATIVE:			
PRESENT	IMPERFECT	PRETERITE	FUTURE
canto	cantaba	canté	cantaré
cantas	cantabas	cantaste	cantarás
canta	cantaba	cantó	cantará
cantamos	cantábamos	cantamos	cantaremos
cantáis	cantabais	cantasteis	cantaréis
cantan	cantaban	cantaron	cantarán
CONDITIONAL:			
cantaría			
cantarías			
cantaría			
cantaríamos			
cantaríais			
cantarían			
SUBJUNCTIVE:			
PRESENT	IMPERFECT-1	IMPERFECT-2	FUTURE
cante	cantara	cantase	cantare
cantes	cantaras	cantases	cantares
cante	cantara	cantase	cantare
cantemos	cantáramos	cantásemos	cantáremos
cantéis	cantaraís	cantaseis	cantareis
canten	cantaran	cantasen	cantaren
IMPERATIVE:			
canta			
cantad			
INFINITIVE:	GERUND:	PAST PARTICIPLE:	
cantar	cantando	cantado	

Given the feature system in (13), the pervasive neutralization of the distinction between the 2nd and 3rd conjugations can then be captured by the following Impoverishment Rules. Note that the designation $[+\text{Pres}, -\text{Fin}]_X$ in rule (16) holds of at least the gerund suffix and the *-nte* adjective-forming suffix in Spanish.

$$(15) \quad +\beta \rightarrow \emptyset / [__]_{TH} [-\text{Pres}]_{TAM}$$

$$(16) \quad +\beta \rightarrow \emptyset / [__]_{TH} [+ \text{Pres}, -\text{Fin}]_X$$

Neither (15) nor (16) applies in the present subjunctive, a fact that deserves some comment given that the 2nd and 3rd conjugations look identical in this part of the paradigm too.

I follow Harris (1969, pp.98-100) in proposing that the *a* in e.g. *coma* ‘eat.3sg.pres.sbjv’ and *viva* ‘live.3sg.pres.sbjv’ is not a manifestation of the Theme Vowel, but rather of what is labeled in this paper as TAM (see also Harris 1987, p.76 for an additional argument for this segmentation). The Theme Vowel in each case is underlyingly the same as in the present indicative, and regular phonology gives /kom-e-a/ $\Rightarrow$ [koma] and /bib-i-a/ $\Rightarrow$ [biβa] respectively, where underlining indicates floating material. I follow Wetzels (1995) on Portuguese in proposing that rules of floating segment association apply from right to left in Spanish. As a result, in a sequence of floating segments “competing” for a single timing slot, the rightmost such segment will be the one that surfaces (a fuller sample derivation illustrating this is provided in section 4.3).

(17) **2nd conjugation verb *comer* ‘to eat’**

INDICATIVE:			
PRESENT	IMPERFECT	PRETERITE	FUTURE
como	comía	comí	comeré
comes	comías	comiste	comerás
come	comía	comió	comerá
comemos	comíamos	comimos	comeremos
coméis	comíais	comisteis	comeréis
comen	comían	comieron	comerán
CONDITIONAL:			
comería			
comerías			
comería			
comeríamos			
comeríais			
comerían			
SUBJUNCTIVE:			
PRESENT	IMPERFECT-1	IMPERFECT-2	FUTURE
coma	comiera	comiese	comiere
comas	comieras	comieses	comieres
coma	comiera	comiese	comiere
comamos	comiéramos	comiésemos	comiéremos
comáis	comierais	comieseis	comiereis
coman	comieran	comiesen	comieren
IMPERATIVE:			
come			
comed			
INFINITIVE:	GERUND:	PAST PARTICIPLE:	
comer	comiendo	comido	

Another notable fact about rules (15) and (16) is that, while these two rules delete the same feature, they cannot be collapsed. This is because Impoverishment Rules must be stated over natural classes. One might legitimately ask whether this fact should be seen as a weakness of the account. I believe that the answer is ‘no’. The reason is that Italian has an equivalent of rule (16) (*vendere* ‘to sell’ /*vendendo* ‘sell.ger’; *dormire* ‘to sleep’ /*dormendo* ‘sleep.ger’), but

no equivalent of rule (15) (*vendevo* ‘I sell.impfv’ vs *dormivo* ‘I sleep.impfv’). This supports the treatment of these as two separate neutralizations: Italian shows us that you can have one but not the other (note in passing that the markedness relationship between the 2nd and 3rd conjugations is reversed in Italian relative to Spanish: in Italian, the distinction is neutralized in favor of the 2nd conjugation, at least in the gerund).

(18) **3rd conjugation verb *vivir* ‘to live’**

INDICATIVE:			
PRESENT	IMPERFECT	PRETERITE	FUTURE
vivo	vivía	viví	viviré
vives	vivías	viviste	vivirás
vive	vivía	vivió	vivirá
vivimos	vivíamos	vivimos	viviremos
vivís	vivíais	vivisteis	viviréis
viven	vivían	vivieron	vivirán
CONDITIONAL:			
viviría			
vivirías			
viviría			
viviríamos			
viviríais			
vivirían			
SUBJUNCTIVE:			
PRESENT	IMPERFECT-1	IMPERFECT-2	FUTURE
viva	viviera	viviese	viviere
vivas	vivieras	vivieses	vivieres
viva	viviera	viviese	viviere
vivamos	viviéramos	viviésemos	viviéremos
viváis	vivierais	vivieseis	viviereis
vivan	vivieran	viviesen	vivieren
IMPERATIVE:			
vive			
vivid			
INFINITIVE:			
vivir	viviendo	PAST PARTICIPLE:	
		vivid	

Finally, note that (15) applies in the imperfect indicative (we have *comer/comía* alongside *vivir/vivía*), meaning that it must apply before the Impoverishment Rule in (9). This counter-bleeding order might not need to be stipulated; it would follow automatically if Impoverishment Rules are subject to the Cyclic Principle (see Middleton 2023). This is because the TH node targeted by (15) is lower in the structure than the TAM node targeted by (9). After these rules have applied, the resulting structure is the input to Vocabulary Insertion. The Rules of Exponence for the TH node are as shown in (19).

- (19) a. TH $\leftrightarrow$ a
b. TH_{+α,-β} $\leftrightarrow$ e
c. TH_{+α} $\leftrightarrow$ I

The significance of the underlining here has already been explained. Otherwise, the exponents introduced by rules (19a) and (19b) have their usual IPA values, but the one in (19c) (a capital ‘i’, meant to indicate a representationally special version of /i/) requires further comment.

Most varieties of Spanish famously have five surface vowels: [a, e, i, o, u]. Now, most instances of these surface vowels undergo no important alternations. In my analysis, these non-alternating vowels reflect underlying /i/, /e/, /a/, /o/, and /u/, which are fully specified for all features and are pre-linked to the Timing Tier. However, at least the four [-Low] vowels also have variants that undergo certain seemingly morphologically restricted alternations. These are summarized in (20).

(20) **Special Vowel Alternations**

- a. In many roots of the third conjugation, [i] alternates with [e] depending on the height of the following vowel (*pedimos* ‘we request’ vs *pidamos* ‘we request.sbjv’).
- b. The Theme Vowel allomorph introduced by Rule (19c) alternates between [e] and [i] depending on the position of stress (*pedir* ‘to request’ vs *pide* ‘(s)he requests’).
- c. The Theme Vowel allomorph introduced by Rule (19c) also undergoes diphthongization when stressed and followed by certain combinations of segments (see below).
- d. In many roots, [e] alternates with [je] when stressed (*querer* ‘to want’ vs *quiero* ‘I want’).
- e. In many roots, [o] alternates with [we] when stressed (*contar* ‘to count’ vs *cuento* ‘I count’).
- f. In a handful of verbs of the third conjugation, [i] alternates with [je] when stressed (*adquirir* ‘to acquire’ vs *adquiero* ‘I acquire’).
- g. In a small handful of verbs of the third conjugation, [i] alternates with both [je] and [e], in part depending on the position of stress (*mentir* ‘to lie (in the sense of deceive)’ vs *miento* ‘I lie’ vs *mintió* ‘(s)he lied’).
- h. In exactly one verb root, [u] alternates with [we] when stressed. This is *jugar* ‘to play’; cf *juega* ‘(s)he plays’.
- i. In a tiny number of third conjugation verb roots, [u] alternates with [o] and with [we]; again, stress seems to be involved (*dormir* ‘to sleep’ vs *duerme* ‘(s)he sleeps’ vs *durmió* ‘(s)he slept’).

Different grammatical theories differ markedly in their analysis of such non-automatic, morphophonological alternations. There is enough evidence from acquisition, historical change, and psycholinguistic experimentation to convince me that the patterns exhibited by such semi-irregular forms have a life of their own in speakers’ grammars, foreclosing an analysis that writes them off as suppletion and leaves things at that (here I agree with Benz 2022; Embick & Halle 2005). In addition, I would like to avoid resurrecting the *Via Rules* of Vennemann (1972), so for me a suppletion analysis is foreclosed entirely (though see Bermúdez-Otero 2022 for an analysis that pursues this option). At the same time, however, the desire to maintain the strongest possible version of modularity has led me away from wanting to pursue an analysis in terms of Indexed Constraints (Pater 2000, 2006, 2008; Gouskova 2012; many others) or Readjustment Rules (for various criticisms of Readjustment, see amongst others Bermúdez-Otero 2012, 2013, 2016; Haugen 2016; Haugen & Siddiqi 2016; Newell & Sailor 2023; Pérez Herrera 2023, p.12; Scheer 2011, 2012, 2020; it seems to me that Indexed Constraints raise similar modularity-related concerns). Therefore, the only option left open to me is to pursue a phonological account of the observations

in (20). That is, alternating /i/s must be underlyingly different in some way from non-alternating /i/s, and likewise *mutatis mutandis* for the other alternations.

Fortunately, the tools to build such an analysis are at hand, in the form of (i) Strict CV as a model of the Timing Tier (Lowenstamm 1996; Scheer 2004; many others) and (ii) underspecification in URs as way of dealing with not-fully-automatic alternations (Inkelas 1996), and in particular the implementation of this idea found in recent work in the theory now known as Logical Phonology (Hale & Reiss 2008; Bale & Reiss 2018; Dabbous et al. 2025; Gorman & Reiss 2024). To begin the discussion, I repeat rule (19c), and supply Rules of Exponence for a sample of interesting 3rd conjugation verb roots, in (21).

- (21) a. $\text{TH}_{+\alpha} \leftrightarrow \underline{\text{I}}$
b. $\sqrt{\text{live}} \leftrightarrow \text{viv}$ (vivir, vive, vivamos, vivieron)
c. $\sqrt{\text{merse}} \leftrightarrow \text{merx}$ (sumergir, sumerege, sumerjamos, sumergieron)
d. $\sqrt{\text{cern}} \leftrightarrow \text{se}^{\text{CV}}\text{rn}$ (discernir, discierne, discernamos, discernieron)
e. $\sqrt{\text{request}} \leftrightarrow \text{pId}$ (pedir, pide, pidamos, pidieron)
f. $\sqrt{\text{serve}} \leftrightarrow \text{sIrb}$ (servir, sirve, sirvamos, sirvieron)
g. $\sqrt{\text{boil}} \leftrightarrow \text{I}^{\text{CV}}\text{rb}$ (hervir, hierve, hirvamos, hirvieron)
h. $\sqrt{\text{sleep}} \leftrightarrow \text{dU}^{\text{CV}}\text{rm}$ (dormir, duerme, durmamos, durmieron)
i. $\sqrt{\text{quire}} \leftrightarrow \text{ki}^{\text{CV}}\text{r}$ (adquirir, adquiere, adquiramos, adquirieron)

In (21), /I/ and /U/ signify vowels which are underspecified for the feature [+/- high], but are specified for the feature [-low] and for roundness and backness (with /I/ being unrounded and front, and /U/ being rounded and back). As discussed earlier, a superscript ‘CV’ indicates that the vowel is accompanied by an additional empty CV sequence on the Timing Tier, and underlining indicates that the segment is not linked to the Timing Tier underlyingly (here, this is relevant only for the Theme Vowel in (21a)).⁹ These URs will be mapped to the appropriate surface representations by rules introduced in section 4.3; before that, though, it is necessary to introduce Logical Phonology and Strict CV in somewhat more detail.

4.2 The Melodic Tier and the Timing Tier

4.2.1 The Theory of the Melodic Tier: Logical Phonology, Underspecification, and Unification

The Melodic Tier consists of linearly ordered bundles of features. I adopt the traditional idea that segments are sets of binary features. Furthermore, segments need not be specified for all features

⁹There turn out to be striking patterns amongst the 3rd conjugation roots (see especially Albright 2008; Albright et al. 2001; Brame & Bordelois 1973, 1974). Specifically, it seems that the presence or absence of a diphthong alternation may be at least partly predictable from the consonant(s) at the end of a 3rd conjugation root, in a way not entirely at variance with what I propose below for the 3rd conjugation Theme Vowel. While I do not wish to rule out the possibility that these patterns might be appropriately represented in the phonological rule system itself, I have not tried to do this here (see especially Harris 1974, 1980 for a skeptical evaluation of the prospect). The results from Albright and colleagues indicate that the generalizations must be represented somehow, however—if not in the phonological rule system, then perhaps in the form of sublexica over the Rules of Exponence for such roots, in the sense of Gouskova et al. 2015; Gouskova 2025; Tabachnick 2023, and many others. Given the present account’s eschewal of Readjustment Rules and similar devices, calculations over sublexica would have to feed into the choice of the UR (specifically, whether a new form looks like it belongs to the set of existing forms whose UR contains underspecified material or empty timing slots of the relevant sort), rather than the assignment of readjustment-triggering diacritics, as in the works just cited.

(Inkelas 1996). Both of these tenets are fundamental to Logical Phonology (Hale & Reiss 2008; Bale & Reiss 2018; Dabbous et al. 2025; Gorman & Reiss 2024).

Like almost all phonological theories with which I am familiar, Logical Phonology acknowledges a qualitative distinction between *automatic* and *non-automatic* alternations (the latter being those that seem to be morphologically restricted in some way). The specific appeal of Logical Phonology in this context is that the theory makes it possible to analyze both types of alternation in the narrow phonology without compromising strong modularity. Let us see how this is so.

In Logical Phonology, the traditional notion of structural change associated with Rule-Based Phonology (conventionally symbolized as ‘ $\rightarrow$ ’) is broken down into more basic operations inspired by formal logic. The most important of these for present purposes is *Unification*, symbolized by ‘ $\sqcup$ ’. It is defined by Dabbous et al. (2025, p.8) as follows:

(22) **Unification**

If A and B are feature specifications, and their union $A \cup B$ is consistent, then $A \sqcup B = A \cup B$; otherwise, $A \sqcup B = A$.

Automatic alternations in quality involve a pairing of a Unification rule with a Subtraction Rule. Subtraction Rules involve set subtraction (symbolized by ‘ $\setminus$ ’) over distinctive features (Dabbous et al. 2025, p.7).

(23) **Set Subtraction**

$x \in X \setminus Y$ iff $x \in X$ and $x \notin Y$.

Hence, a fully productive rule voicing obstruents after nasals, which would be written as (24) in traditional Rule-Based Phonology, might, in a Logical Phonology analysis, consist of the Subtraction Rule in (25) followed by the Unification rule in (26).

(24) $[+obstruent] \rightarrow +voice / +nasal ______$ (Traditional Chomsky & Halle 1968-style rule)

(25) $[+obstruent] \setminus \{-voice\} / +nasal ______$ (Subtraction)

(26) $[+obstruent] \sqcup \{+voice\}$ (Unification)

Non-Automatic Alternations instead involve a conjunction of a special, underspecified Underlying Representation and a Unification Rule. For example, given the Underlying Representation for the root of the verb *pedir* ‘to request’ provided in (21e), the following Unification Rules from Gorman & Reiss (2024) will yield forms in *ped* when the following vowel is [i], and forms in *pid* otherwise (the necessary rule ordering here is given by the Elsewhere Condition; note that I will slightly reformulate (27) below).

(27) $\left[\begin{array}{c} -back \\ -low \end{array} \right] \sqcup \{-high\} / ______ C_0 \left[\begin{array}{c} -back \\ +high \end{array} \right]$

(28) $[-low] \sqcup \{+high\}$

All non-alternating [e]s and [i]s will be specified for the feature $[+/-high]$ underlyingly, so that the rules in (27) and (28) will apply to them vacuously.

A learner must establish whether a given alternation is Automatic or not, presumably by means of the Tolerance Principle of Yang (2016). In the event that the Tolerance Threshold is exceeded, the learner will conclude that the alternation is Non-Automatic, and will proceed to construct an abstract underlying form by taking the intersection of the features of the surface alternants, minus the ones with inconsistent values (Reiss has referred to this operation as *Generalized Intersection*;

see for instance Unit 45 in Bale & Reiss 2018). For example, in the case of a verb like *pedir*, the underlying /I/ of the root will be arrived at from the two surface alternants [e] and [i], in the manner shown:

$$(29) \quad \begin{bmatrix} -\text{low} \\ -\text{high} \\ -\text{back} \\ -\text{round} \end{bmatrix}_e \cap_G \begin{bmatrix} -\text{low} \\ +\text{high} \\ -\text{back} \\ -\text{round} \end{bmatrix}_i = \begin{bmatrix} -\text{low} \\ -\text{back} \\ -\text{round} \end{bmatrix}_I$$

This much is sufficient background regarding the theory of the Melodic Tier adopted in this paper. Next, we will turn to the Timing Tier.

4.2.2 The Theory of the Timing Tier: Strict CV

Strict CV (see e.g. Lowenstamm 1996; Scheer 2004) is a version of Autosegmental Phonology (Goldsmith 1976) in which the Timing Tier consists only of repeated CV units. In such a model, notions like ‘onset’, ‘nucleus’, and ‘coda’ cannot be defined in terms of constituent structure as in more familiar theories of the syllable. Instead, such notions can only be reconstructed derivatively in terms of lateral relationships between C slots and V slots, and whether they are filled or not. For example, a ‘Nucleus’ is a filled V slot. ‘Coda’ consonants generally correspond to a C with a filled V somewhere before it (immediately before it if it’s the first or only consonant in the coda, further away if it’s later in a complex coda), but an unfilled V immediately after it. A ‘Simple Onset’ is a C with a filled V slot immediately to its right. ‘Complex Onsets’ only arise under special circumstances which will not be germane here.

In Strict CV, all facts which in other theories are dealt with in terms of constraints on constituents of syllable structure also have to be recast in terms of restrictions on when V and C slots are allowed to be unfilled, what they can be filled by under what circumstances, and so on. This is implemented in terms of two lateral relations which can hold between C and V slots: *Government* and *Licensing*. Government and Licensing have in common that they proceed from right to left, that only filled timing slots can initiate them, and that they are mutually exclusive (no timing slot can be simultaneously Governed and Licensed). In situations where both Government and Licensing could apply to a given item, Government takes priority. Filled V slots can Govern and License things on their left. Filled C slots cannot License, and they can Govern only in a special circumstance called *Infrasegmental Government*, but this concept will not play a role in the account to come. Most important here will be ungoverned empty V slots. These, given my adoption of the updated version of the analysis in Harris (1985) developed by Pérez Herrera (2023), will drive the explanation of diphthongization.

The stress system plays a crucial role in all analyses of the Spanish verb conjugation, and the present approach is no exception. Accordingly, I will conclude this subsection by introducing the analysis of Spanish verb stress presupposed in section 4.3. Here we encounter an area in which a small retreat from strict modularity is necessary: Spanish verb stress is copiously studied, and almost everyone agrees some amount of at least indirect reference to morphosyntactic structure is necessary in any analysis of it (a small sample of the literature illustrating this: Doner 2017; Halle et al. 1991; Harris, 1969, 1980, 1985, 1987; Oltra-Massuet & Arregi 2005, Roca, 2005, 2020). The form that this will take in the present analysis is as follows: as part of the transduction from syntactic structure to phonology, alongside the Rules of Exponence, a choice is made about the set of stress-related rules to be employed. Likewise, particular cyclic domains may be special in the set of Timing Tier-manipulating rules they do and don’t invoke; this is a rule-based implementation of a limited form of the Cophonologies by Ph(r)ase approach (Sande et al., 2020).

I take it that stress assignment takes place in two steps: the first is called Accent Assignment, which identifies a specific filled V slot as metrically prominent. The second is called CV-Infixation (Larsen, 1998), which inserts an empty CV sequence to the immediate right of the accented syllable. Since these are two different steps, in principle other processes can intervene between them, and this will be critical to the analysis to come. Furthermore, their separateness raises the possibility that they could be applicable to different cyclic domains. Indeed, I will propose below that, while Accent Assignment takes place on every Cycle, CV-Infixation crucially does not take place during the cycle defined by v^*/Voice .

As for how Accent Assignment itself takes place, I adapt the proposal of Doner (2017), which is itself an extension of the system in Roca (2005). This analysis employs the fundamentals of the Halle/Idsardi metrical system (Idsardi 1992; Halle & Idsardi 1995), but extends it to allow for edge marking parameters on Line 0. These edge marking parameters may vary in different morphosyntactic environments. Here is the algorithm as presented by Doner (2017, p.242); note I have omitted a constraint on foot construction on Line 1 (since it plays no role in my analysis).

(30) Line 0

- a. Edge: XXR (Put an X bracket to the X side of the rightmost element)
- b. Iterative Constituent Construction: L (insert a left bracket after every 2nd element proceeding leftwards)
- c. Head: L (the head is on the left)

(31) Line 1

- a. Edge: RRR
- b. Head: R
- c. Conflation: Line 1

In addition to the value of X in the Line 0 Edge-marking rule, Doner (2017, p.243, her Table 2) also allows for variation in whether the Edge Marking Rule applies to the whole word (corresponding to the domain parameter setting ‘W’), or to the word minus the person/number endings (notated using the domain parameter setting ‘S’ for ‘stem’). I will depart from Doner in this respect, replacing this domain parameter with a specification in the Underlying Representation of certain affixes, represented by a single curly bracket ‘}’. I will also allow the URs to contain lexically projecting ‘(’ brackets, for cases that seem to call for underlyingly accented endings (including the 1st singular and 3rd singular preterite indicative endings, respectively $-\acute{e}/-í$ and $-\acute{o}$, which are stressed in verbs with regular preterites).

With these ideas in place, the Edge Marking Rules for Spanish verbs can be simplified from Doner’s discussion to the following:

(32) **Edge Marking Rules for Spanish Verb Conjugation**

- a. **Default:** RRR
- b. **Special:** LLR (infinitive, plural positive imperative)

To simplify the discussion in section 4.3, the result of Accent Assignment will simply be indicated there by bolding and underlining the accented V slot, rather than showing the full calculation for each case. Some sample calculations are presented below for illustration; in each of these, the bracket inserted by Edge Marking on Line 0 is bolded.

- (33) Accent Assignment for the form *pidieron* ‘they request.pret.indic’:
Line 0 Edge Marking RRR

Line 1: *)

Line 0: * (* *)

C	V	C	V	C	V	C	V
p	I	d	I	r	o	n	

- (34) Accent Assignment for the form *pedir* ‘to request’:
Line 0 Edge Marking LLR

Line 1: *)

Line 0: * (*

C	V	C	V	C	V
p	I	d	I	r	

- (35) Accent Assignment for the form *pedíamos* ‘we request.ipfv.indic’:
Line 0 Edge Marking RRR, with special Edge Marking Domain boundary

Line 1: *)

Line 0: * (* *)} *

C	V	C	V	C	V	C	V	C	V
p	I	d	I		a	m	o	s	

- (36) Accent Assignment for the form *pidió* ‘s/he request.pret.indic’:
Line 0 Edge Marking RRR, with Underlying ‘(’ boundary (highlighted)

Line 2: *

Line 1: * *)

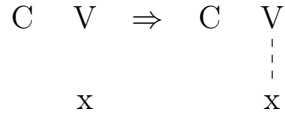
Line 0: (* * (*)

C	V	C	V	C	V
p	I	d	I		o

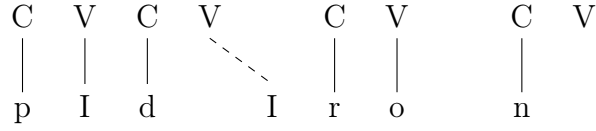
4.2.3 Interim Summary

This subsection has set out some essential technical details regarding the theories of the Melodic Tier and of the Timing Tier presupposed by the phonological analysis to come. Both Logical

(39) **Floating Vowel Association (proceeds right to left)**



(40) ***Pidieron* Step 1: Floating Vowel Association**



The right-to-left directionality of this association rule is not crucial to this form, but see the discussion of *pidamos* ‘we request.pres.sbjv’ below.

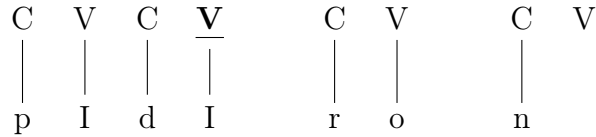
The output of Floating Vowel Association is the input to the rule in (41).

(41) **Accent Assignment**

“Calculate which V slot bears accent.”

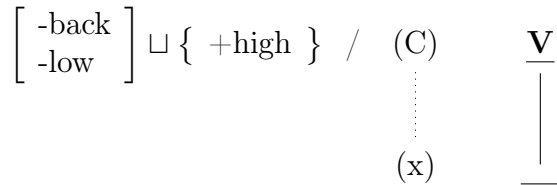
(See (30), (31), (32), and the sample calculations in section 4.2.2)

(42) ***Pidieron* Step 2: Accent Assignment**

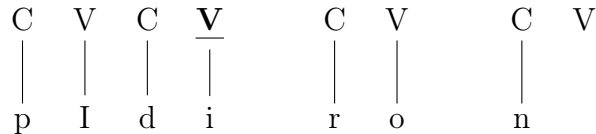


It turns out that the vowel represented by /I/ never surfaces as [e] when accented. When it carries the accent, /I/ always surfaces either as [i] or as [je]. The rule in (43) accounts for the first of these outputs.

(43) **/I/ becomes [i] when accented**



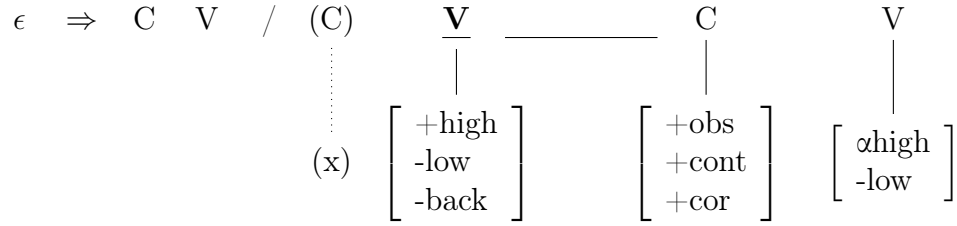
(44) ***Pidieron* Step 3: /I/ becomes [i] when accented**



The diphthongal alternant of accented /I/ is derived by having the output of rule (43) be the input to a rule of *Accented [i] Lengthening*. This rule, which is partly based on proposals in Foley (1965, pp.91-100), Harris (1969, p.186, his (25)), and Brame & Bordelois (1973), has the subcases in (45) and (46) (these could be collapsed by employing LSLT/SPE’s brace notation over the conditioning environments; I have avoided this for typographical and legibility-related reasons).

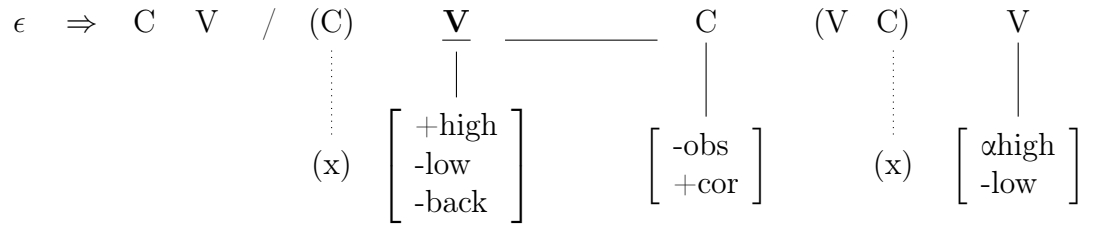
(45) **Accented [i] Lengthening, Subcase 1**

“Insert a CV sequence immediately after an accented V slot linked to a [+high, –low, –back] melody, immediately preceding a C slot filled by a coronal fricative followed immediately by a V slot filled by a [–low] melody which is specified for [+ / – high].”



(46) **Accented [i] Lengthening, Subcase 2**

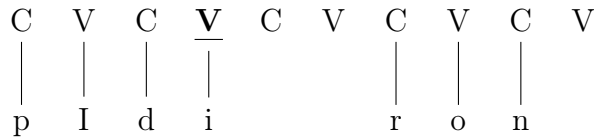
“Insert a CV sequence immediately after an accented V slot linked to a [+high, –low, –back] melody, immediately preceding a C slot filled by a coronal sonorant such that the next filled V slot to its right has a [–low] melody and is specified for [+ / – high].”



Note that there are no ellipses on the Melodic Tier in subcases (45) and (46). This means that the presence of floating material on the Melodic Tier intervening between the critical following C and the next filled vowel slot to its right will prevent Accented [i] Lengthening from applying.¹⁰

Returning to the derivation of *pidieron*, Subcase 2 of Accented [i] Lengthening applies to (44) to yield the representation in (47):

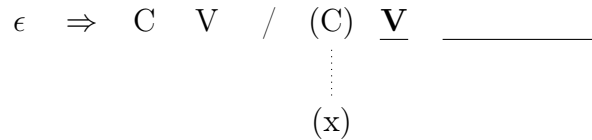
(47) ***Pidieron* Step 4: Accented [i] Lengthening**



Following Larsen (1998), the phonological effects of stress are produced by infixing an empty CV sequence immediately after the accented V slot, via the following rule:

(48) **CV Infixation** (Larsen 1998)

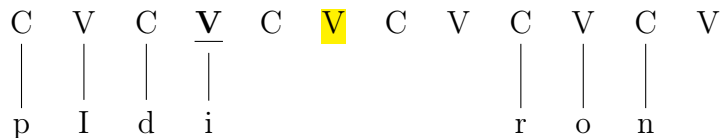
“Insert an empty CV sequence to the immediate right of the accented V slot”



¹⁰This accounts for why e.g. underlying /pis-a-o/ ‘I step (on something)’ surfaces as [piso] rather than *[pjeso]: the presence of the floating /a/ on the timing tier makes Accented [i] Lengthening inapplicable.

Applying (48) to (47) gives us (49).

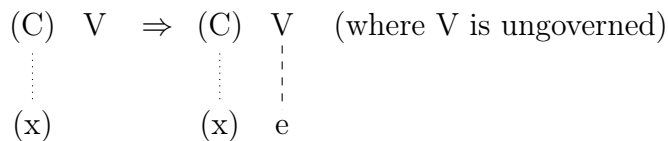
(49) **Pidieron Step 5: CV Infixation**



The highlighted ‘V’ slot in (49) is ungoverned, because the next V slot to its right is empty. Adopting the account in Pérez Herrera (2023), which is itself an updating of the account in Harris (1985), this state of affairs will trigger the rule of Epenthesis of [e] in (50), which will in turn trigger the rule of Accent Displacement in (51), which will in turn trigger the Gliding rule in (52).

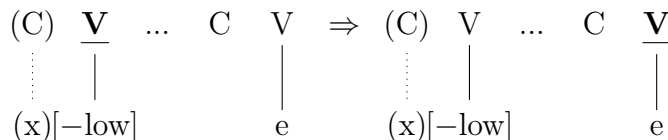
(50) **Epenthesis** (Applies left-to-right)¹¹

“Insert [e] into an ungoverned ‘V’ Slot.”



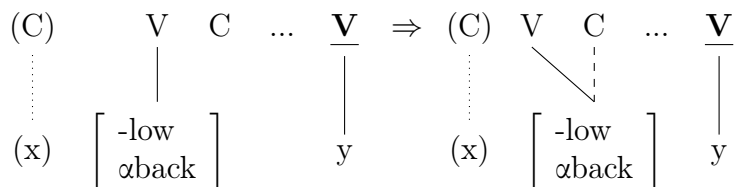
(51) **Accent Displacement**

“If an accented non-low vowel is followed by a ‘V’ slot filled by [e], and no filled C or V slots intervene, move the accent onto the [e].”

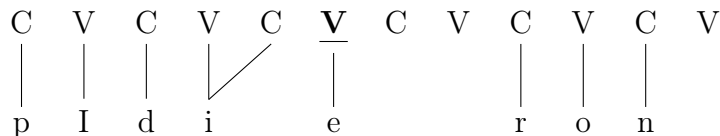


(52) **Gliding**

“Associate a linked non-low vowel to a following empty ‘C’ slot, if the next filled ‘V’ slot is accented, and no filled C slot intervenes.”



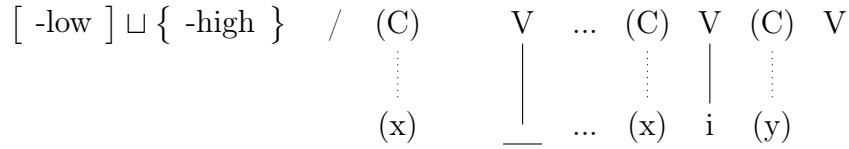
(53) **Pidieron Steps 6,7,8: Epenthesis, Accent Displacement, Gliding**



¹¹The fact that epenthesis applies left-to-right in Spanish is shown by the fact that initial /sC/ clusters have epenthesis to the left of the /s/, rather than to the immediate left of /C/; English *snob* was borrowed as [esnɒb], not as *[senɒb].

The next two rules to consider are in (54) (itself a slight reformulation of a rule given by Gorman & Reiss 2024, reproduced earlier in (27)) and (55). The conditioning environment of (54) is crucially not met by (53), with the result that the less specific rule in (55) will apply instead.

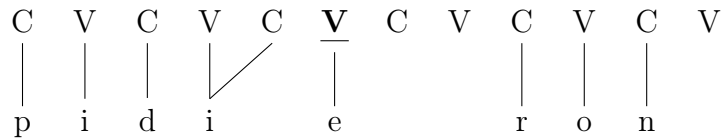
- (54) **/I/ and /U^(CV)/ become [e] and [o] when the next linked nucleus is an [i] which is not shared with a following C slot**



- (55) **Elsewhere, /I/ and /U^(CV)/ become [i] and [u] respectively**
 $[-\text{low}] \sqcup \{+\text{high}\}$

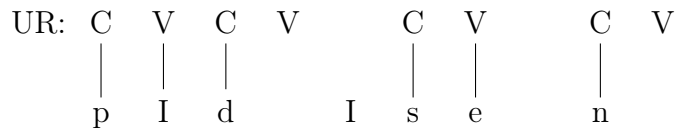
With this step, the derivation of *pidieron* ‘they requested.pret.indic’ is complete. The output (modulo additional rules deriving an automatic spirantizing alternation that affects voiced stops) is the representation in (56).

- (56) ***Pidieron* Step 9 and Output: Rule (55)**

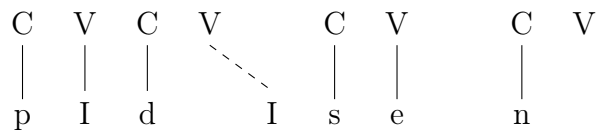


The derivation of *pidiesen* is identical to the one we have just seen, with the exception that it is Subcase 1 of Accented [i] Lengthening that is relevant to this example.

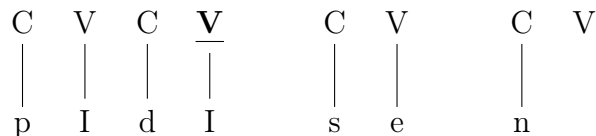
- (57) Syntax: $[[[[\sqrt{\text{request}} \ v] \ \text{TH}_{+\alpha}] \ \text{TAM}_{ipfv.sbjv2}] \ \text{Agr:3pl}]$



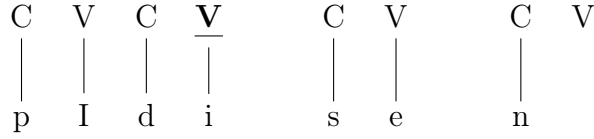
- (58) ***Pidiesen* Step 1: Floating Vowel Association**



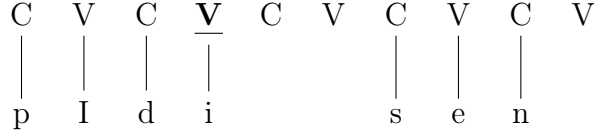
- (59) ***Pidiesen* Step 2: Accent Assignment**



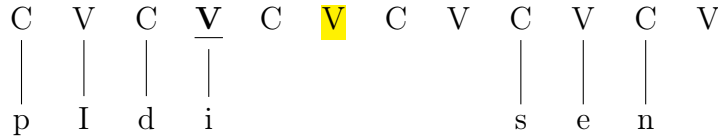
(60) ***Pidiesen* Step 3: /I/ becomes [i] when accented**



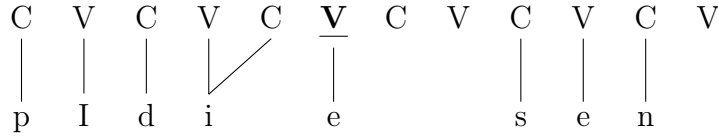
(61) ***Pidiesen* Step 4: Accented [i] Lengthening**



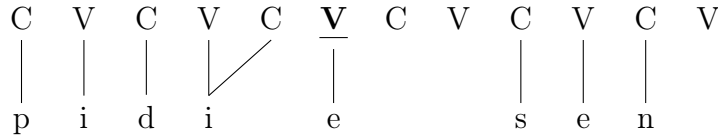
(62) ***Pidiesen* Step 5: CV Infixation**



(63) ***Pidiesen* Steps 6,7,8: Epenthesis, Accent Displacement, Gliding**

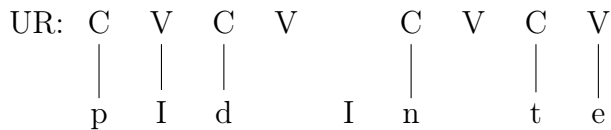


(64) ***Pidiesen* Step 9 and Output: Unification Rule (55) applies to I**

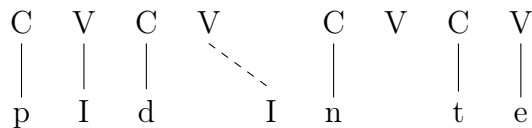


For *pidiente* ‘one who requests (derived adjective/noun)’ the rules involved are the same as for *pidieron*, with one addition. The instructive aspect of the form *pidiente* lies in motivating the additional optional material in the conditioning environment of Subcase 2 of Accented [i] Lengthening, in illustrating that Epenthesis may be triggered *before* CV-Infixation, and in having a role for the rule of Empty VC deletion (Gussmann & Kaye 1993), given below as (74).

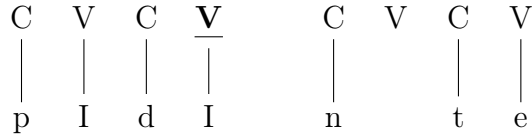
(65) Syntax: [[[$\sqrt{\text{request}}$ v] TH₊ α] Adj]



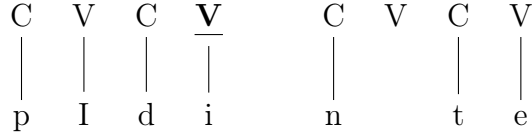
(66) ***Pidiente* Step 1: Floating Vowel Association**



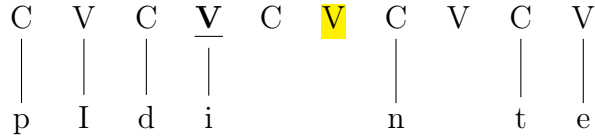
(67) ***Pidiente* Step 2: Accent Assignment**



(68) ***Pidiente* Step 3: /i/ becomes [i] when Accented**

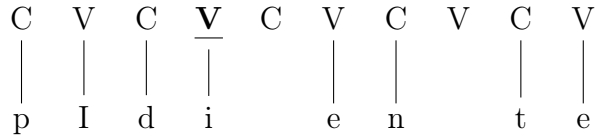


(69) ***Pidiente* Step 4: Accented [i] Lengthening**



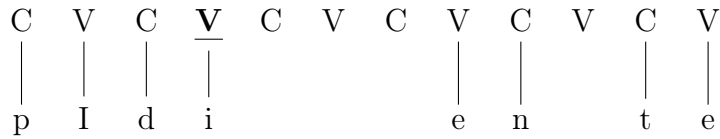
At this point, a new complication arises, owing to the phonological shape of the adjectivalizing affix *-nte*: the output of Stressed [i] Lengthening contains an ungoverned vowel. I assume that this causes epenthesis to apply at once, with the result seen in (70).

(70) ***Pidiente* Step 5: Epenthesis**



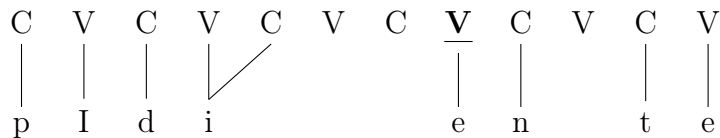
Next, CV infixation applies.

(71) ***Pidiente* Step 6: CV Infixation**



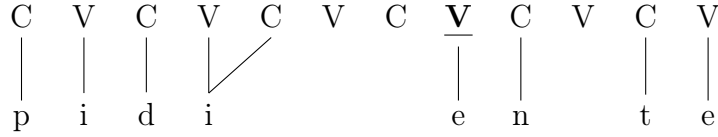
Next, the Rule of Accent Displacement will apply, switching the accent onto the epenthetic [e]. The new configuration will meet the structural description for Gliding, which will duly apply as well.

(72) ***Pidiente* Steps 7+8: Accent Displacement and Gliding**



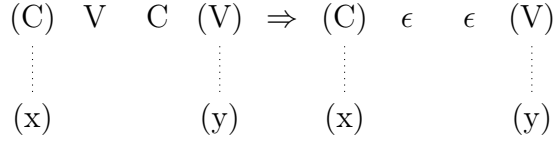
The penultimate step is the Unification Rule which makes underspecified vowels [+high].

(73) **Pidente Step 9: Unification Rule (55) applies to I**

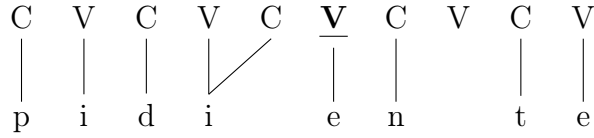


Finally, the rule of Empty VC Reduction in (74) will apply, giving the output in (75).

(74) **Empty VC Reduction** (Gussmann & Kaye 1993, p. 433)

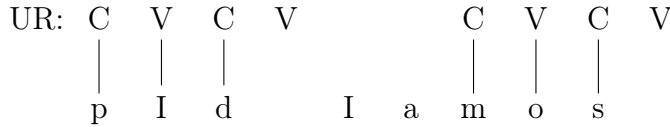


(75) **Pidente Step 10 and Output: Empty VC Reduction**

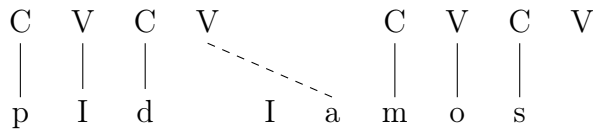


The form *pidamos* ‘we request.pres.sbjv’ is quite straightforwardly derived. Its sample derivation is included here to illustrate the crucial role played by the assumption that Floating Vowel Association proceeds from right to left.

(76) Syntax: [[[[$\sqrt{\text{request}}$ v] TH₊ α] TAM_{pres.sbjv}] Agr:1pl]

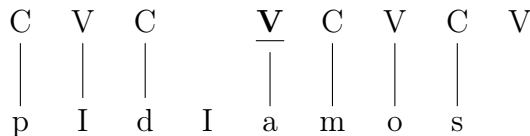


(77) **Pidamos Step 1: Floating Vowel Association**

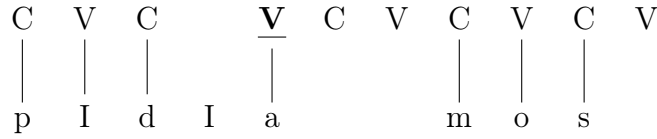


As we can see in (77), the fact that Floating Vowel Association applies from right-to-left means that the Theme Vowel never gets to link to the Timing Tier in this derivation. Accent Assignment applies next, followed by CV-Infixation.

(78) **Pidamos Step 2: Accent Assignment**

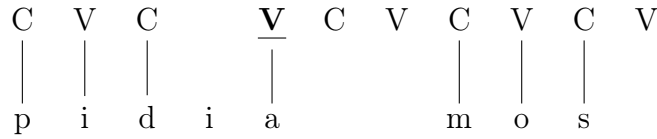


(79) ***Pidamos* Step 3: CV-Infixation**



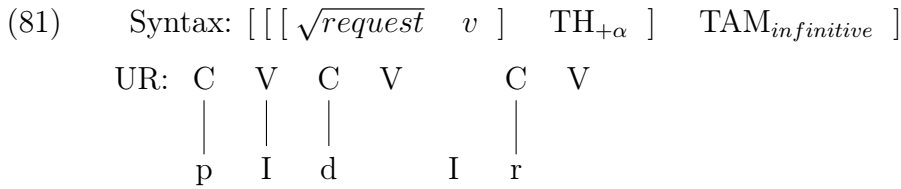
Unlike in the first three sample derivations, the structural description of Accented [i] Lengthening is never met here, and the application of CV-Infixation in (79) has not created any ungoverned V slots. Hence, the chain reaction that leads to surface diphthongization is not triggered. Unification Rule (55) applies to the /I/ of the root (and, vacuously, to that of the Theme Vowel). We thus have, in (80), the output *modulo* spirantization.

(80) ***Pidamos* Step 4 and Output: Unification Rule (55)**



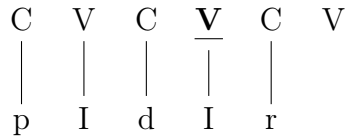
The form *pedirían* ‘they would request’ involves a more complex derivation, which highlights a crucial role for cyclicity in the account, as well as an additional case where a certain amount of coarse-grained reference to morphosyntactic structure is needed for certain rules which interact with the Timing Tier.

The first cycle of this derivation involves the syntax and UR below.



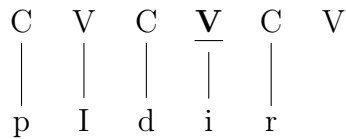
Floating Vowel Association and Accent Assignment apply in that order to yield (82).

(82) ***Pedirían* Cycle 1, Steps 1+2: FVA and Accent Assignment**



The next step is the application of rule (43).

(83) ***Pedirían* Cycle 1, Step 3: Accented /I/ becomes [i]**



The structural description of Accented [i] Lengthening is not met by this representation. Although the structural description of CV-Infixation *is* met at this point, I propose that **this rule never applies on the vP cycle** (this idea will, I believe, also play a role in enabling the present approach to sidestep the so-called *Problem of the Missing Cycle* in a manner similar to Halle et al. 1991).¹² The next rule whose structural description is met is the Unification Rule assigning [–high] when the next non-glide V slot is filled by [i].

(84) ***Pedirían* Cycle 1, Step 4: Rule (54)**

C	V	C	<u>V</u>	C	V
p	e	d	i	r	

This concludes the First Cycle. The start of the second cycle will look as follows:

(85) ***Pedirían* Cycle 2: Input**

C	V	C	<u>V</u>	C	V	C	V	C	V	C	V
p	e	d	i	r		I		a		n	

Floating Vowel Association, crucially ordered before Epenthesis, will apply, followed by a cycle of the Accent Assignment rules, which will result in the newly linked Theme Vowel being accented.

(86) ***Pedirían* Cycle 2 Steps 1+2: FVA and Accent Assignment**

C	V	C	V	C	V	C	<u>V</u>	C	V	C	V
p	e	d	i	r		I		a		n	

The only remaining rules whose Structural Descriptions are met are (43) and CV-infixation, which will apply here (this cycle is defined by CP, not vP).

(87) ***Pedirían* Cycle 2 Steps 3+4: Accented /i/ becomes [i]; CV-infixation**

C	V	C	V	C	V	C	<u>V</u>	C	V	C	V	C	V
p	e	d	i	r		i				a		n	

CV-Infixation does not give rise to any ungoverned vowels, meaning that Epenthesis is not triggered. The representation in (87) will form the input to empty VC-reduction, after the which the automatic spirantization rules will yield the output.

¹²For somewhat more recent discussion of the Problem of the Missing Cycle generally, see Bermúdez-Otero (2013) (where the problem is given its name), Myler (2015), and Bermúdez-Otero (2016). I agree with Bermúdez-Otero (2016) that the account in Myler (2015) is inadequate in various respects; I hope to return in future work to the question of whether the conjecture in the text will fix the inadequacies in Myler's account.

The remaining sample derivations illustrate how the highly similar URs for the roots of the verbs *hervir* ‘to boil’ and *servir* ‘to serve’, respectively /I^{CV}rb/ and /sIrb/, interact with the rule system to give rise to the alternations seen in the forms *hierve* ‘it boils.pres.indic’, *hirvió* ‘it boiled.pret.indic’, *sirve* ‘it serves.pres.indic’, and *sirvió* ‘it served.pret.indic’. Note that, in all these forms, I assume that the coda /r/ is also linked to the V slot following it, making it representationally a glide. As well as preventing what would otherwise be a source of over-generation involving epenthesis, this hypothesis also seems like the most straightforward way of accounting for the fact that voiced stops that follow /r/ undergo spirantization.

$$(88) \quad \text{Syntax: } \left[\left[\left[\left[\sqrt{\text{boil}} \quad v \right] \quad \text{TH}_{+\alpha} \right] \quad \text{TAM}_{\text{pres.indic}} \right] \quad \text{Agr:3sg} \right]$$

$$\text{UR: } \begin{array}{ccccccc} \text{C} & \text{V} & \text{C} & \text{V} & \text{C} & \text{V} & \text{C} & \text{V} \\ & | & & & | & / & | & \\ & \text{I} & & & \text{r} & & \text{b} & \text{I} \end{array}$$

(89) ***hierve* Steps 1+2: FVA and Accent Assignment**

$$\begin{array}{ccccccc} \text{C} & \underline{\text{V}} & \text{C} & \text{V} & \text{C} & \text{V} & \text{C} & \text{V} \\ & | & & & | & / & | & \text{---} \\ & \text{I} & & & \text{r} & & \text{b} & \text{I} \end{array}$$

(90) ***hierve* Step 3: Accented /I/ becomes [i]**

$$\begin{array}{ccccccc} \text{C} & \underline{\text{V}} & \text{C} & \text{V} & \text{C} & \text{V} & \text{C} & \text{V} \\ & | & & & | & / & | & | \\ & \text{i} & & & \text{r} & & \text{b} & \text{I} \end{array}$$

Note that the structural description of Accented [i] Lengthening is not met in (90): the next pure vocalic nucleus is filled by /I/, and /I/ has no height specification. The next rule to apply is therefore CV-Infixation, which will trigger the by-now-familiar chain reaction that leads to diphthongization.

(91) ***hierve* Step 4: CV-Infixation**

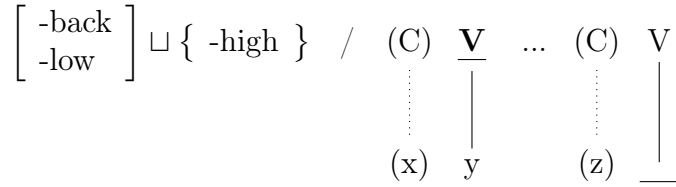
$$\begin{array}{ccccccc} \text{C} & \underline{\text{V}} & \text{C} & \text{V} & \text{C} & \text{V} & \text{C} & \text{V} \\ & | & & & | & / & | & | \\ & \text{i} & & & \text{r} & & \text{b} & \text{I} \end{array}$$

(92) ***hierve* Steps 5,6,7: Epenthesis, Accent Displacement, Gliding**

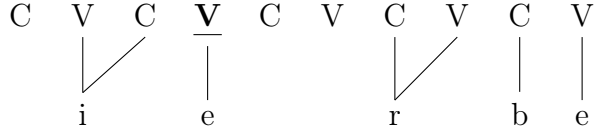
$$\begin{array}{ccccccc} \text{C} & \text{V} & \text{C} & \underline{\text{V}} & \text{C} & \text{V} & \text{C} & \text{V} \\ & | & / & | & & & | & | \\ & \text{i} & & \text{e} & & & \text{r} & \text{b} & \text{I} \end{array}$$

At this point, an additional rule becomes relevant. This one, which is loosely based on Harris (1969, p.98, his (9)), has the effect of making /I/ into [e] when it follows the accented syllable.

(93) /I/ becomes [e] when post-tonic

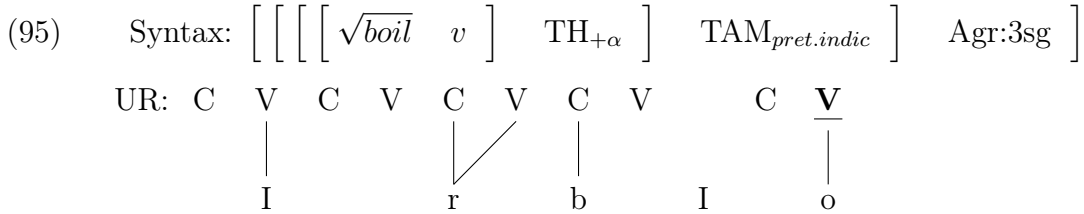


(94) *hervve* Step 8: Rule (93)



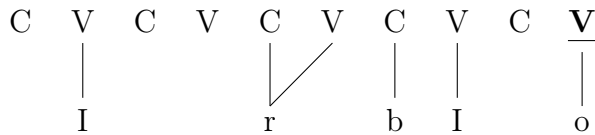
The representation in (94) forms the input to spirantization, yielding the output [jerβε].

The derivation of the form *hirvió* provides us with our first opportunity to observe the effect of an underlyingly accented morpheme (i.e., one that projects a left parenthesis onto Line 0). This arises in the realization of the preterite indicative TAM head in the environment of a 3rd person singular subject: (*o*).



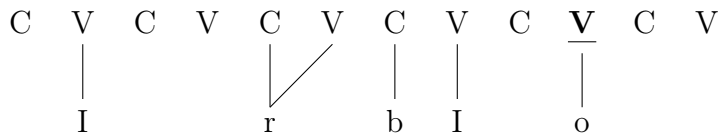
Floating Vowel Association and Accent Assignment have the following result:

(96) *hirvió* Steps 1+2: FVA and Accent Assignment



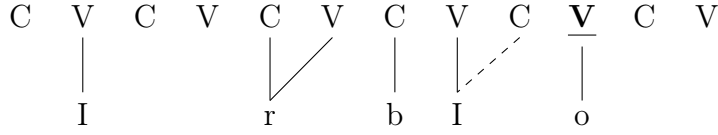
When CV-infixation applies, the empty V-slot it inserts will be domain-final, meaning that it is governed (no empty V slot immediately follows it).

(97) *hirvió* Step 3: CV-Infixation



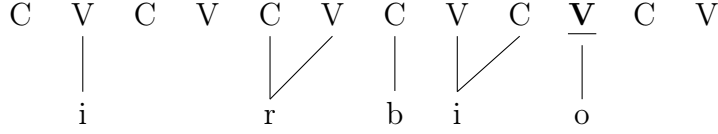
Since there are no ungoverned V slots in (97), Epenthesis does not apply. Nevertheless, the rule of Gliding in (52) will apply here.

(98) ***hirvió* Step 4: Gliding**



Rule (55) will then set the height value of both /I/'s to '+':

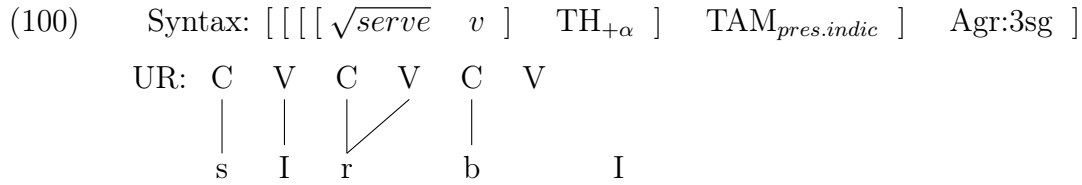
(99) ***hirvió* Step 5: (55)**



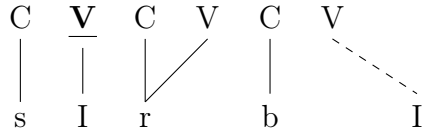
Modulo spirantization, (99) is the desired output.

The derivations for *sirve* 'it serves.pres.indic' and *sirvió* 'it served.pret.indic' are straightforward, and are therefore presented with little comment. The key component that explains why this verb behaves the same as *hervir* 'to boil' in the preterite but differently in the present is the absence in the root for 'serve' of an empty CV sequence following the vowel in the UR of the root.

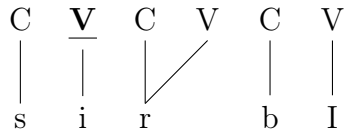
First we have the derivation for *sirve* 'it serves.pres.indic':



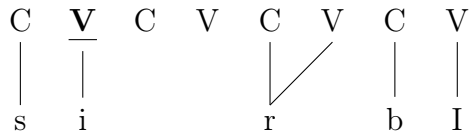
(101) ***Sirve* Steps 1+2: FVA and Accent Assignment**



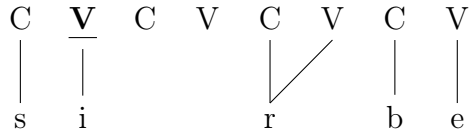
(102) ***Sirve* Step 3: Accented /I/ becomes [i]**



(103) ***Sirve* Step 4: CV-Infixation**

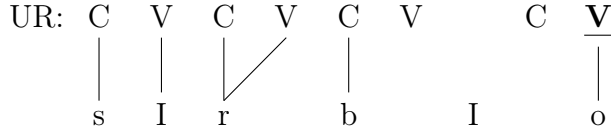


- (104) ***Sirve* Step 5 (and output, *modulo* Spirantization):**
Post-Tonic /I/ becomes [e]

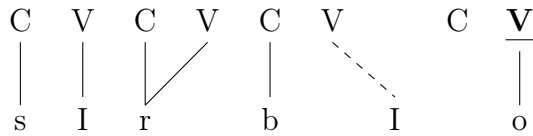


The derivation for *sirvió* ‘it served.pret.indic’ proceeds as follows:

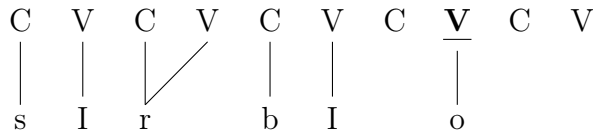
- (105) Syntax: [[[[$\sqrt{\text{serve}}$ v] TH₊ α] TAM_{pret.indic}] Agr:3sg]



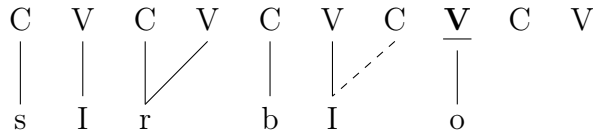
- (106) ***Sirvió* Steps 1+2: FVA and Accent Assignment**



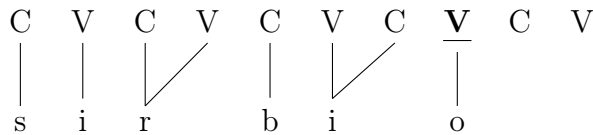
- (107) ***Sirvió* Step 3: CV-Infixation**



- (108) ***Sirvió* Step 4: Gliding**



- (109) ***Sirvió* Step 5 (and output, *modulo* Spirantization): Rule (55)**



4.4 Summary on (Theme) Vowel Alternations

This section has introduced phonological rules which cover a significant portion of the non-automatic vowel alternations in Spanish. This system derives the vowel alternation-related aspects of the PYTA pattern as a special case, as summarized in (110).

(110) **Explanation of the Vowel Alternations which produce (and go beyond!) PYTA**

- a. The underlying vowel /I/ undergoes diphthongization when stressed and followed by specific combinations of segments (see (45) and (46)).
- b. The underlying vowel /I/ surfaces as [i] when stressed in all environments not covered by (110a).
- c. The underlying vowel /I/ surfaces as [e] when post-tonic.
- d. The underlying vowels /I/ and /U/ surface as [−high] when they are unstressed and the next vowel to the right is [i].
- e. The underlying vowels /I/ and /U/ surface as [+high] when they are unstressed elsewhere.

5 Conclusion

I began this paper by highlighting two questions on which Realizational approaches to morphology continue to be divided.

(111) **A Big Question (about pieces)**

Is the input to the Rules of Exponence (a) a syntactic (or at least syntax-like) hierarchical structure, as on piece-based views; or (b) an unstructured bag of features, as on word-based views?

(112) **Another Big Question (about paradigms)**

Are Rules of Exponence (a) sensitive only to information in the structure they are exponing, or are they also (b) potentially sensitive to the way other structures are realized, via transderivational principles that regulate form correspondences?

The core contribution of this paper has been to show that the PYTA Morpheme, at least as it manifests itself in Spanish, cannot be cited as forcing the (b) answers. Once one has accounted independently for the suppletive stem allomorphy and non-automatic vowel alternations that characterize Spanish (and one must account for these independently, since they each spread in different ways outside the claimed morphomic pattern), there is no work left over for a direct morphomic statement to do. This conclusion is quite independent of the specific hypotheses I have put forward to account for these generalizations, which are summarized as follows:

(113) **Two Interlocking Patterns that Give Rise to PYTA**

- a. Relevant suppletive allomorphs are conditioned by a coherent morphosyntactic environment: [−Pres], usually in combination with [+Fin].
- b. Theme Vowel Diphthongization and Root-internal height alternations are both governed by phonology.

Since both of these components are important in explaining forms outside the PYTA pattern as well as within it, they are independently motivated, rather than being ad-hoc devices whose only role is to explain away the morphomic pattern. The broader research program implied by this should be clear: it involves surveying claimed instances of morphemes and showing that they are all either (i) piece-based, like the Latin *t*-stem, or (ii) spurious, like PYTA in Spanish.

Even if the reader refuses to accept my specific analysis, and even if they feel that the implied research program I have just outlined is doomed to failure, I think they will find it impossible to avoid the following conclusion. Looking across available approaches to process-based morphomic patterns, it seems that we all have a choice between one of the following three conclusions.

(114) **Here are Three Bullets, Please Bite at Least One**

- a. (Morpho)phonology and Exponence involve transderivational comparison of outputs.
- b. Phonology is not a module in the strict Fodorian sense: rules (and/or constraints) that manipulate phonological representations can make direct reference to morphosyntactic primes of all/most kinds.
- c. Phonology allows for Underlying Representations which are not limited to representing contrastive features, and for processes which are indifferent or even inimical to phonetic markedness.

Bullet (114a) is the one offered to us by word-and-paradigm approaches, and by certain imaginable Optimality Theoretic accounts (perhaps based on the one in Steriade 2016). Bullet (114b) corresponds to accounts involving Indexed Constraints or Readjustment Rules. Finally, Bullet (114c) is the one offered by the present paper: this involves forsaking a leading idea about the nature of phonology (that it is fundamentally about the endless conflict between contrast and markedness, and how this conflict is resolved in particular cases) in favor of an entirely different one; one in which phonology is both abstract and substance-free. This, as far as I can tell, exhausts the existing families of approach to process-based morphemes in the literature. The reader may be able to imagine others. I, as I write these words, cannot. In any case, I think it is unlikely that any additional alternative will sidestep the following conclusion: the battle over process-based morphemes will be a battle for the very soul of phonology.

6 Appendices

6.1 Appendix 1: The Pieces

$$(115) \quad \begin{array}{l} \text{1st Conjugation Root} \\ \left[\begin{array}{l} \text{CAT: Root, ...} \\ \text{CLASS:} -\alpha \end{array} \right]_{\text{Root1}} \end{array}$$

$$(116) \quad \begin{array}{l} \text{2nd Conjugation Root} \\ \left[\begin{array}{l} \text{CAT: Root, ...} \\ \text{CLASS:} +\alpha, +\beta \end{array} \right]_{\text{Root2}} \end{array}$$

$$(117) \quad \begin{array}{l} \text{3rd Conjugation Root} \\ \left[\begin{array}{l} \text{CAT: Root, ...} \\ \text{CLASS:} +\alpha \end{array} \right]_{\text{Root3}} \end{array}$$

$$(118) \quad \begin{array}{l} \text{Root-attaching Verbalizer} \\ \left[\begin{array}{l} \text{CAT: v} \\ \bullet \text{Root} \\ \bullet \text{TH} \\ \dots \end{array} \right]_v \end{array}$$

$$(119) \quad \begin{array}{l} \text{Theme Vowel} \\ \left[\begin{array}{l} \text{CAT: TH} \\ \square \text{Class} \end{array} \right]_{TH} \end{array}$$

$$(120) \quad \begin{array}{l} \text{Present Indicative} \\ \left[\begin{array}{l} \text{CAT:TAM} \\ \bullet \text{v} \\ +\text{PRES} \\ -\text{PERF} \\ -\text{SBJV} \\ +\text{FIN} \end{array} \right]_{\text{pres.indic}} \end{array}$$

$$(121) \quad \begin{array}{l} \text{Present Subjunctive} \\ \left[\begin{array}{l} \text{CAT:TAM} \\ \bullet \text{v} \\ +\text{PRES} \\ -\text{PERF} \\ +\text{SBJV} \\ +\text{FIN} \end{array} \right]_{\text{pres.sbjv}} \end{array}$$

$$(122) \quad \begin{array}{l} \text{Preterite Indicative} \\ \left[\begin{array}{l} \text{CAT:TAM} \\ \bullet \text{v} \\ -\text{PRES} \\ +\text{PAST} \\ +\text{PERF} \\ -\text{SBJV} \\ +\text{FIN} \end{array} \right]_{\text{pret.indic}} \end{array}$$

$$(123) \quad \text{Imperfect Indicative} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ -\text{PRES} \\ +\text{PAST} \\ -\text{PERF} \\ -\text{SBJV} \\ +\text{FIN} \end{array} \right]_{\text{imper.f.indic}}$$

$$(124) \quad \text{Imperfect Subjunctive 1} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ -\text{PRES} \\ +\text{PAST} \\ -\text{PERF} \\ +\text{SBJV} \\ +\text{FIN} \\ +1 \end{array} \right]_{\text{imper.f.sbjv1}}$$

$$(125) \quad \text{Imperfect Subjunctive 2} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ -\text{PRES} \\ +\text{PAST} \\ -\text{PERF} \\ +\text{SBJV} \\ +\text{FIN} \\ -1 \end{array} \right]_{\text{imper.f.sbjv2}}$$

$$(126) \quad \text{Future Subjunctive} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ -\text{PRES} \\ -\text{PAST} \\ +\text{SBJV} \\ +\text{FIN} \end{array} \right]_{\text{fut.sbjv}}$$

$$(127) \quad \text{Infinitive} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ -\text{FIN} \\ +\text{INFIN} \end{array} \right]_{\text{infinitive}}$$

$$(128) \quad \text{Gerund} \\ \left[\begin{array}{c} \text{CAT:TAM} \\ \bullet \mathbf{v} \\ +\text{PRES} \\ -\text{FIN} \\ -\text{INFIN} \end{array} \right]_{\text{ger}}$$

- (129) Past Participle

$$\left[\begin{array}{l} \text{CAT:TAM} \\ \bullet v \\ -\text{PRES} \\ -\text{FIN} \\ -\text{INFIN} \end{array} \right]_{\text{past.part}}$$
- (130) Subject Agreement

$$\left[\begin{array}{l} \text{CAT:AGR} \\ \bullet \text{TAM}_{+\text{fin}} \\ \square \text{Pers} \\ \square \text{Num} \end{array} \right]_{\text{Agr}}$$
- (131) Future/Conditional Builder

$$\left[\begin{array}{l} \text{CAT: } v, \text{ fut/cond} \\ \text{CLASS: } +\alpha, +\beta \\ \bullet \text{TAM}_{+\text{infin}} \\ \bullet \text{TH} \\ \text{oTAM}_{-\text{perf}, -\text{sbjv}} \end{array} \right]_{v \text{ fut/cond}}$$

6.2 Appendix 2: Rules of Impoverishment

- (132) $+\beta \rightarrow \emptyset / [__]_{TH} [-\text{Pres}]_{TAM}$
- (133) $+\beta \rightarrow \emptyset / [__]_{TH} [+Pres, -Fin]_X$
- (134) $-\text{Pres} \rightarrow \emptyset / [__, +\text{Past}, -\text{Perf}, -\text{Sbjv}]_{TAM}$

6.3 Appendix 3: Sample Rules of Exponence

- (135) **Roots**
- a. $\sqrt{\text{want}} \leftrightarrow \text{kis} / __ \dots [-Pres, +Fin]_{TAM}$
 - b. $\sqrt{\text{want}} \leftrightarrow \text{ke}^{CV} \text{r}$
 - c. $\sqrt{\text{live}} \leftrightarrow \text{viv} (\text{vivir}, \text{vive}, \text{vivamos}, \text{vivieron})$
 - d. $\sqrt{\text{merse}} \leftrightarrow \text{merx} (\text{sumergir}, \text{sumerge}, \text{sumerjamos}, \text{sumergieron})$
 - e. $\sqrt{\text{cern}} \leftrightarrow \text{se}^{CV} \text{rn} (\text{discernir}, \text{discierne}, \text{discernamos}, \text{discernieron})$
 - f. $\sqrt{\text{request}} \leftrightarrow \text{pId} (\text{pedir}, \text{pide}, \text{pidamos}, \text{pidieron})$
 - g. $\sqrt{\text{serve}} \leftrightarrow \text{sIrb} (\text{servir}, \text{sirve}, \text{sirvamos}, \text{sirvieron})$
 - h. $\sqrt{\text{boil}} \leftrightarrow \text{I}^{CV} \text{rb} (\text{hervir}, \text{hierve}, \text{hirvamos}, \text{hirvieron})$
 - i. $\sqrt{\text{sleep}} \leftrightarrow \text{dU}^{CV} \text{rm} (\text{dormir}, \text{duerme}, \text{durmamos}, \text{durmieron})$
 - j. $\sqrt{\text{quire}} \leftrightarrow \text{ki}^{CV} \text{r} (\text{adquirir}, \text{adquiere}, \text{adquiramos}, \text{adquirieron})$

(136) **Verbalizers**

- a. $v_{fut/cond} \leftrightarrow (a / _ \dots [+Pres]_{TAM} [-author, -pl]_{AGR}$
- b. $v_{fut/cond} \leftrightarrow (a / _ \dots [+Pres]_{TAM} [-participant, +pl]_{AGR}$
- c. $v_{fut/cond} \leftrightarrow (CV$
- d. $v \leftrightarrow \emptyset$

(137) **Theme Vowels**

- a. $TH \leftrightarrow \underline{a}$
- b. $TH_{+\alpha, -\beta} \leftrightarrow \underline{e}$
- c. $TH_{+\alpha} \leftrightarrow \underline{I}$

(138) **TAM Morphemes and Deverbal Derivational Morphemes**

- a. $Adj \leftrightarrow nte$
- b. $TAM \leftrightarrow \emptyset$
- c. $TAM_{[+pres, +sbjv]} \leftrightarrow \underline{e} / [-\alpha]_{TH} _$
- d. $TAM_{[+pres, +sbjv]} \leftrightarrow \underline{a} / [+ \alpha]_{TH} _$
- e. $TAM_{[+past, -perf, -sbjv]} \leftrightarrow ba\} / [-\alpha]_{TH} _$
- f. $TAM_{[+past, -perf, -sbjv]} \leftrightarrow a\} / [+ \alpha]_{TH} _$
- g. $TAM_{[+past, +perf, -sbjv]} \leftrightarrow (e / [-\alpha]_{TH} _ Agr:1sg$
- h. $TAM_{[+past, +perf, -sbjv]} \leftrightarrow (i / [+ \alpha]_{TH} _ Agr:1sg$
- i. $TAM_{[+past, +perf, -sbjv]} \leftrightarrow ste\} / _ Agr:2$
- j. $TAM_{[+past, +perf, -sbjv]} \leftrightarrow (o / _ Agr:3sg$
- k. $TAM_{[+past, +perf, -sbjv]} \leftrightarrow ro / _ Agr:3pl$
- l. $TAM_{[+past, +sbjv, +1]} \leftrightarrow ra\}$
- m. $TAM_{[+past, +sbjv, -1]} \leftrightarrow se\}$
- n. $TAM_{[-pres, -past, +sbjv]} \leftrightarrow re\}$
- o. $TAM_{[+infin]} \leftrightarrow r_{LLR}$
- p. $TAM_{[-fin, -pres]} \leftrightarrow d$
- q. $TAM_{[-fin, +pres]} \leftrightarrow nd$

(139) **Agr Morphemes**

- a. $Agr:1sg \leftrightarrow \underline{o} / [+pres, -sbjv]_{TAM} _$
- b. $Agr:1sg \leftrightarrow \emptyset$
- c. $Agr:2sg \leftrightarrow \emptyset / [-pres, +past, +perf, -sbjv]_{TAM} _$ ¹³
- d. $Agr:2sg \leftrightarrow s$
- e. $Agr:3sg \leftrightarrow \emptyset$
- f. $Agr:1pl \leftrightarrow mos$
- g. $Agr:2pl \leftrightarrow is$
- h. $Agr:3pl \leftrightarrow n$

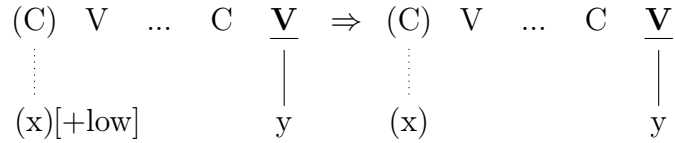
¹³Many varieties of Spanish lack this rule, so that they have e.g. *hablastes* rather than the normative form *hablaste* for ‘You_{sg, informal} spoke’ (the 2nd person singular preterite).

6.4 Appendix 4: Phonological Rules

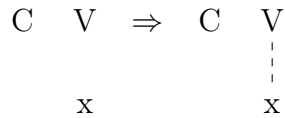
The rule in (140) plays no role in the main text discussion, but it is needed in a full analysis of Spanish to explain such preterite forms as *habló* [abló] ‘(s)he spoke’ from underlying /abl-a-(o/; without rule (140), we would incorrectly predict the form *[ablaó].

(140) **Pretonic Floating Vowel Stripping**

“If a floating low vowel precedes a vowel that has projected a left bracket to Line Zero, and no filled consonant slot intervenes, delete the floating vowel.”



(141) **Floating Vowel Association (proceeds right to left)**



(142) **Accent Assignment**

a. **Edge Marking Rules for Spanish Verb Conjugation**

- i. **Default:** RRR
- ii. **Special:** LLR (infinitive, plural positive imperative)

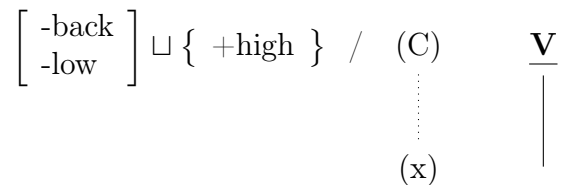
b. **Line 0**

- i. **Edge:** XXR (Put an X bracket to the X side of the rightmost element)
- ii. **Iterative Constituent Construction:** L (insert a left bracket after every 2nd element proceeding leftwards)
- iii. **Head:** L (the head is on the left)

c. **Line 1**

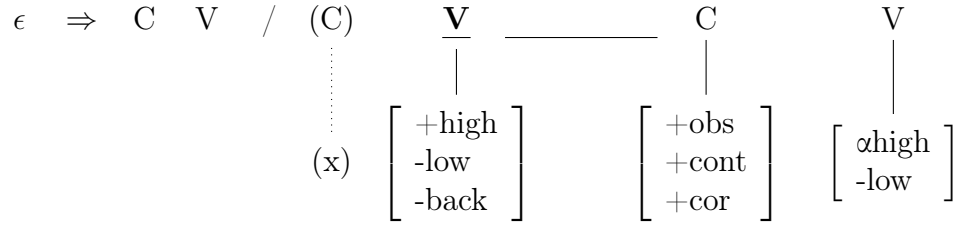
- i. **Edge:** RRR
- ii. **Head:** R
- iii. **Conflation:** Line 1

(143) **/I/ becomes [i] when accented**



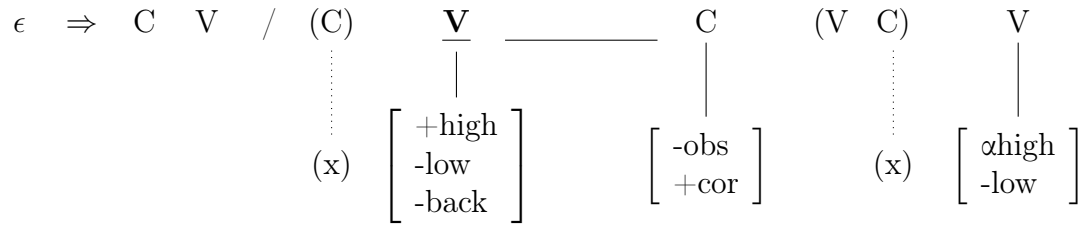
(144) **Accented [i] Lengthening, Subcase 1**

“Insert a CV sequence immediately after an accented V slot linked to a [+high – low, –back] melody, immediately preceding a C slot filled by a coronal fricative followed immediately by a V slot filled by a [–low] melody which is specified for [+ / – high].”



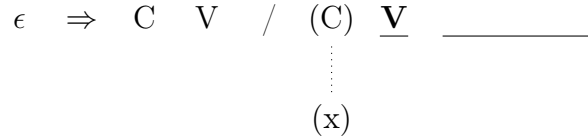
(145) **Accented [i] Lengthening, Subcase 2**

“Insert a CV sequence immediately after an accented V slot linked to a [+high, –low, –back] melody, immediately preceding a C slot filled by a coronal sonorant such that the next filled V slot to its right has a [–low] melody and is specified for [+ / – high].”

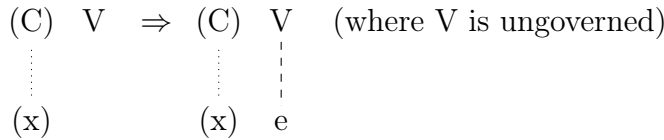


(146) **CV Infixation** (Larsen 1998)

“Insert an empty CV sequence to the immediate right of the accented V slot”

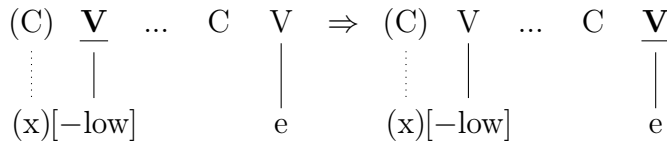


(147) **Epenthesis (proceeds left-to-right)**



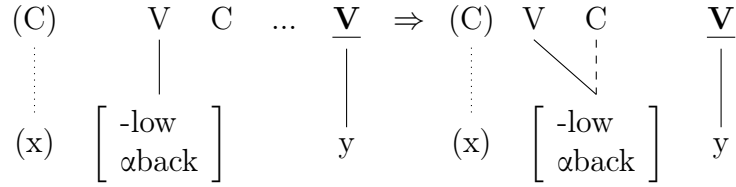
(148) **Accent Displacement**

“If an accented non-low vowel is followed by a ‘V’ slot filled by [e], and no filled C or V slots intervene, move the accent onto the [e].”

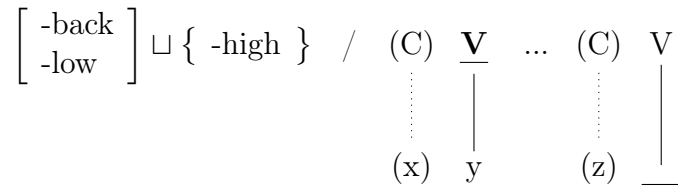


(149) **Gliding**

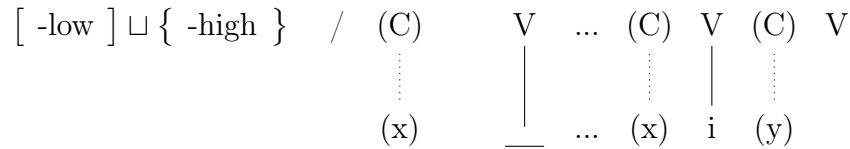
“Associate a linked non-low vowel to a following empty ‘C’ slot, if the next filled ‘V’ slot is accented, and no filled C slot intervenes.”



(150) **/I/ becomes [e] when post-tonic**

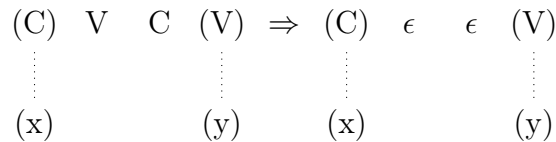


(151) **/I/ and /U^(CV)/ become [e] and [o] when the next linked nucleus is an [i] which is not shared with a following C slot**



(152) **Elsewhere, /I/ and /U^(CV)/ become [i] and [u] respectively**
 $\left[\text{-low} \right] \sqcup \{ \text{+high} \}$

(153) **Empty VC Reduction** (Gussmann & Kaye 1993, p. 433)



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